Reached through the imaginary, real by construction:

the maximally symmetric substrate as the geometric–ontological core of Cosmological Relativity

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Abstract

The papers of this programme construct, read, and apply a single object: the maximally symmetric de Sitter substrate. This paper is about the object itself. We make explicit three facts the corpus has carried implicitly and never stated in one place. First, the substrate is *everywhere intrinsically real by construction*: it is a real manifold with a real coordinate basis—the five-dimensional $\mathrm{dS}_5 = SO(5,1)/SO(4,1)$ of the ladder below, of which the four-dimensional dS_4 is the *background* its leaves carry—whose Lorentzian signature is an intrinsic property of its positive curvature, not a signature imposed on it from outside; the imaginary variables the construction uses to *reach* it—the fifth embedding coordinate, the conjugacy circle, the equatorial seam's analytic continuation—are instruments of visualisation and continuation over a geometry that is real at every point they land on. Second, that substrate is the *universal standard of physics*: its curvature radius $\alpha = \sqrt{3/\Lambda}$ and its locked null cone are the intrinsic length and causal structure against which every material structure is what it is—Eddington's reason the cosmological constant cannot vanish, here corrected (its timelike half as real as its spatial one) and grounded (a real intrinsic ground state, not a property of the operation of measurement); and, read at the last-scattering surface, this same standard dissolves the horizon problem—the microwave background's uniformity is the substrate's maximal symmetry, not the residue of an early causal contact. Third, the *maximal symmetry* of that substrate is a single fact from which the corpus's separately-established results descend as one: the parameter-free rigidity of the cosmology, the fundamental constants standing as unit gauges over the substrate's single scale—a singleness with a consequence the ledger should state, since neither real form supplies a second invariant and a dimensionless magnitude needs two[P17R13]: the global Wick rotation carries the defining quadric to S^5 of the *same* radius, acting on the coordinate and leaving the right-hand side untouched, and every curvature invariant on either face is a pure power of $1/\alpha^2$. *So the construction cannot force a coupling, and its silence about magnitudes is a property of a one-constant theory rather than a gap awaiting work*—the common root of three verdicts reached separately, that the winding quantises without measuring, the flat bundle selects without coupling, and the branch point filters without supplying—, the necessity and sufficiency of the augmentation of general relativity, the universality of physics, and the wall against a continuous geometric matter symmetry are the same property seen at several rungs. Read at that constant rung, the single scale renders the traditional Planck values gauge-combinations rather than physical scales, and dissolves the cosmological-constant and coincidence problems by that one fact—no Planck scale for Λ to be small against, and no bare-Λ-versus-vacuum-energy split for the fine cancellation to act on—with Λ's value the ledger's one input scale, not predicted. We collect the source—a real coordinate basis in which the would-be timelike embedding coordinate is not a coordinate of the manifold at all, and a curvature radius that is the manifold's one scale—and we show that the straight null rulings of the surface, the light cone locked to the equilateral profile that maximal symmetry forces, are the visible peak of exactly this intrinsically real geometry.

That one scale has a classical name. Read projectively, the substrate is a Cayley–Klein[L17] geometry [5, 6] whose *absolute* is the quadric $\eta(X, X) = 0$—the null cone—and its geodesic separation is exactly the Cayley–Klein log-cross-ratio against that absolute, with the Cayley–Klein constant equal to α: timelike, $\tau = (\alpha/2)|\log \mathrm{CR}|$; spacelike, the corresponding angle;

and the discriminant $4\alpha^4(u^2-1)$ in $u = \eta(P,Q)/\alpha^2$ deciding which, according as the joining line meets the absolute or misses it[P17R11]. A second projective fact follows at once, and it settles how the ladder's second rung sits in its first.

Since a point P of the substrate has $\eta(P,P) = \alpha^2 > 0$, its polar hyperplane $\{X : \eta(P,X) = 0\}$ carries signature $(1,4)$, so the substrate cut by it is a totally geodesic dS_4 of isometry $\mathrm{SO}(4,1)$[P17R12].

That subgroup is a sweep-subgroup of the substrate's $\mathrm{SO}(5,1)$, which is the criterion by which a four-geometry is a *cut*; and being the maximally symmetric such cut it is the dS_4 background itself.

So polarity is how the background sits in the substrate—the background is the polar slice of a substrate point, one for each point and none privileged, the point a slice is polar to being what selects it.

The ladder is stated in this paper and the embedding had been stated in none; it is the same pole–polar relation the throat protocol already runs on, read one dimension up. Since the scale is the only free constant a Cayley–Klein construction carries—signature, null structure and isometry group being fixed by the absolute's projective type—the one-scale reading is not merely the outcome of the constants audit but the classical statement about a geometry of this kind.

That null structure has a second face, and it is older than the geometry: read on the substrate's overhead projection, the *power of a point* with respect to the throat—Steiner's invariant, $|X|^2 - \alpha^2$, which for an external point is the square of the tangent length (Euclid III.36)—is, by the hyperboloid's own equation, the square of that point's *height* in the embedding. The tangent from any point of the substrate to the throat therefore runs exactly as far across as the point stands high, and is a *null line*: Euclid's tangent–secant relation and the null condition are the same equation, and the only thing that turns one into the other is the minus sign in the metric. The rulings *are* those tangents, so "doubly ruled by straight null lines" and "every tangent to the throat is null" are one statement, the classical register adding reach—every point of the overhead, not only the ones a ruling passes through. The identity is bounded exactly, and the bound is a statement about the substrate rather than about geometry: it is a fact about *one* circle, the waist, so a classical theorem whose quantity is built from the power with respect to the throat carries the signature for free, while one whose quantity is a chord, a bilinear relation among points, or a power with respect to any other circle carries no height at all. Two ingredients of this are modern and not one: the minus sign, and *power itself*—Euclid proves a relation among segments, and it was Steiner who shifted the focus from the chords to the point, the move this programme makes when it reads the cut's offset not as a coordinate but as the mass. And the reach of that identity is not a mystery to be marvelled at but a consequence of it: the classical theory of the circle speaks about causal structure exactly when it speaks about the power of a point with respect to this one circle. What the reading returns is, in every case, a result the construction already established—the horizon, the light cone, the Nariai configuration [30], the throat's radius, the perspectival reading of the mass—reached by a route sharing no machinery with the one that established it. And the one-circle bound is not finally a limitation but the point: the circle is what the construction is built on.

Its radius *is* the curvature, $\alpha = \sqrt{3/\Lambda}$, the one scale the exhausted isometry leaves; the power vanishes on the circle itself; and the tangents to it are the rulings, so the double ruling and the classical power law are one statement, set by α and by nothing else. That circle is the incircle of the hinge triangle and its nine-point circle besides, so the hinges stand at 2α as an *output* of the hole rather than a stipulation, the three sides are the Nariai double null rulings tangent at their midpoints, and the triple-angle identity returns the Nariai configuration of its own accord. The signed areal radius passes through zero at a point *on* it, so the three walls whose bound modes are the fermion sector's three chiral generations lie on it, their three-foldness the hole's own; and the slicing's lap runs around it, a black hole and a cosmology thereby one object and not two—the cosmological gravitational collapse of black holes in prior universes completing into the formation of universes like our own through the crossing it carries, matter and antimatter the two ends of one conjugation across that crossing.

The lines, the curvature, the placement, the count and the passage are read off the one waist. And the count's own structure is read off it too: the discrete residue is the direct product $\mathrm{Aut}(A_2) = S_3 \times \mathbb{Z}_2$, whose factors act on the three roots and the two rulings—which are, on the waist, the points *on* the circle and the lines *tangent* to it. The residue factorises because those two relations are independent; and the factors differ in *kind* for the same reason, a ruling being a line of the substrate and a root a label of a different cut—so chirality descends gauged and flavour global. The Standard Model's arrangement is, in this reading, the difference between being on the one circle and being tangent to it. All of which is why there is one circle to build on and not a family of them, and which we state as the seventh face of the maximal symmetry this paper's thesis identifies. One consequence of that reading is drawn in §6 and is deliberately *not* counted among the faces: written in a general dimension, the triple angle collapses to a single multiple-angle only in four spacetime dimensions and in five, and the mass-parity that grades chirality exists only in even dimension, so four is the only dimension carrying both a generation count and a chirality. It is excluded from the list because it does not come from maximal symmetry—the substrate is maximally symmetric and moduli-free in *every* dimension—but from the matter sector's own content, and the two are statements about different objects.

It is therefore offered as a second way of seeing the object and not as evidence for it; a reading that returned new physics from a theorem about circles would be an object of suspicion, and one that returns what was already proved, from unrelated directions, is doing what a faithful dictionary does. The frame is a physical theory of beautiful symmetric core structures whose broken symmetries are the physics: the maximally symmetric substrate dS_5, the background dS_4 its leaves carry, and the exact solutions as symmetry-breaking cuts read as shadows off a wall.

The paper is written as the space the rest of the corpus lands in: the slicing curve, the description groupoid, the constraint algebroid, the cosmology, and the matter boundary each descend from this core and return their consolidated content to it.

The core is the seed the corpus grew from and the hub it returns to: the papers descend from it and consolidate their content back into it, the connections their propagation reveals gathered in as they are worked.

1 Introduction: the object, not its uses

Across the corpus one geometry does all the work. The event horizon is a metric singularity of it [12]; the Schwarzschild circle and the Schwarzschild–de Sitter slicing curve are curves drawn on it [13, 14]; the description groupoid charts it from many vantages [15]; the cosmic foliation is measured to be its own congruence [16]; the constraint algebra of general relativity is its symmetric-space grading [24]; the flat-ΛCDM cosmology is its clock [25]; the boundary at which local structure yields to the cosmic expansion—the Hubble–Eddington radius [10]—is the flat locus of its slice, one scale reaching both ranges [14, 25]; and the Standard-Model boundary is the limit of what its symmetry can carry [26]. Each paper builds or reads or applies the substrate. None of them is *about the substrate itself*—about what kind of object it is, and why the two things most easily misread about it are in fact one thing.

This paper is about the object. Its claim has three faces that turn out to be one:

- **Real by construction (§2, §4).** The substrate is a real manifold with a real coordinate basis—dS_5, the ladder's top rung, not the four-dimensional background dS_4 its leaves carry. Its Lorentzian signature is intrinsic to its positive curvature. The imaginary variables the construction reaches it through are bookkeeping over an everywhere-real geometry, never a rotation into a different, unreal space.
- **The universal standard of physics (§3).** Its curvature radius $\alpha = \sqrt{3/\Lambda}$ and its locked null cone are the intrinsic length and causal structure against which every material structure is what it is—the vacuum's "measuring stick throughout space and time." This is Eddington's reason the cosmological constant cannot vanish, corrected and grounded: the universality of physics *is* the maximal symmetry, one real standard the same at every point. The standard is legible in the expansion's own dynamics as well. On a comoving

worldline $\mathrm{d}^2 r/\mathrm{d}\tilde{\tau}^2 = rK_G$ with $K_G = 1/\alpha^2 - M/r^3$ [14]: *the substrate's standard and the material structure enter the acceleration as its two terms, with opposite signs*, and the expansion turns from decelerating to accelerating exactly where the material structure's contribution falls to the substrate's. Set $\Lambda = 0$ and the first term is gone, the acceleration is $-M/r^2$ throughout, and there is no turn—so the sign change is the standard's own signature in the dynamics, which is Eddington's point read where he did not read it. And that causal structure is legible in Euclid's own quantity (§5.1): the power of a point with respect to the throat *is* the square of its height, so the tangent from any point of the substrate to the throat is a null line, and the classical theory of the power of a point is—on this one circle—a theory of the substrate's causal structure. Which bounds it exactly: the identity is a fact about *one* circle, the waist, and a classical theorem is blind to the signature unless its own quantity is built from that power. A second bound is drawn, and it separates this identity from the classical relations that accompany it. Written at a general dimension the hinge figure is a regular $(D-1)$-gon with the throat as its incircle, and of the equivalent statements that hold at the hinge, two survive and three do not. The tangent length equalling the hinge's height survives, because both are $\sqrt{R^2 - \alpha^2}$ whatever the vertex distance R; so does the sides' touching the throat at their midpoints, which is a property of any regular polygon's incircle. What does not survive is the placement $R = 2\alpha$, the 60° subtense, and the identification of the throat with the triangle's nine-point circle—the last not because a number changes but because Euler's nine-point circle is a theorem about triangles and has no analogue at a square. *So the identity this section rests on is the substrate's, and the classical relations that arrive with it at four dimensions are the equilateral triangle's* [14]. They coincide there, which is why they read as one fact.

- **Maximally symmetric, and that is one fact worn several ways (§6).** The rigidity that lets the cosmology hide nowhere against the data, the geometric relations that lock the constants, the necessity and sufficiency of the augmentation, the universality of physics, and the wall against a continuous geometric matter symmetry are the maximal symmetry of the substrate seen at different rungs. The recursion the corpus runs on closes on a fixed ontological core, and the core is maximal symmetry itself.

These are the same fact because the object that is real by construction *is* the object maximal symmetry forces *is* the object that serves as the universal standard: de Sitter is the unique real maximally symmetric manifold whose Lorentzian signature is intrinsic (§2). So every imaginary instrument used to reach it, and every free parameter the constants might have carried, is disposed of by the one property—maximal symmetry leaves the geometry real everywhere, leaves it nothing free to tune, and makes it the one standard everywhere. The three faces are separately established below; identifying them as one fact is this paper's central hypothesis.

The picture this opens onto, and the frame the paper is written in, is a physical theory of beautiful symmetric core structures whose broken symmetries are the physics. There is a ladder of symmetry: the maximally symmetric *substrate* $\mathrm{dS}_5 = SO(5,1)/SO(4,1)$; the maximally symmetric four-dimensional *background* dS_4 its leaves carry; and, below these, the *exact solutions*—Schwarzschild, Nariai, the Friedmann geometries—each a symmetry-breaking cut of the one substrate, matter the bend no symmetry holds in place. What we observe as gravitation, mass, and cosmic history are the manifestations of those broken symmetries, read as shadows off a wall (§7)—perspectival descriptions of one intrinsically real, maximally symmetric object. The geometric core is the standard; the physics is what the standard looks like when its symmetry is cut.

A word on why this is its own paper and not a remark in an existing one. The result that fixes it—the real coordinate basis in which the would-be timelike embedding coordinate is not a coordinate at all—predates the programme; it is proved in the 2012 dissertation whose chapter three this paper takes up at source [1]. The corpus carries the seam-side instance of it as a *guard* [14, 26], and rediscovered the null-ruling face of it as a mechanism [25]. But the general statement—that the whole substrate, seam and cosmogenesis and the phase structure at the seam included, is everywhere intrinsically real, and that this reality is the same fact as its maximal symmetry—has never been made in one place. It is load-bearing enough, and connected to enough, to be the geometric–ontological heart the applications hang from.

2 The substrate is a real manifold

2.1 The maximally symmetric manifolds, intrinsically

The maximally symmetric solutions of $R_{\mu\nu} = \Lambda g_{\mu\nu}$—the geometries isotropic about every point—are, up to the sign and magnitude of Λ, a one-parameter family with curvature radius $\alpha \equiv \sqrt{3/|\Lambda|}$ [1, §sec_RPT]. They are usually met as hypersurfaces embedded in a flat five-dimensional space,

$$\sum_{\mu=0}^{4} x_\mu^2 = \alpha^2, \qquad ds^2 = \sum_{\mu=0}^{4} dx_\mu^2, \tag{1}$$

which is a convenient picture but not the manifold. Solving (1) for the fifth coordinate and substituting back gives the *intrinsic* four-dimensional line element,

$$ds^2 = d\mathbf{x}^2 + \frac{(\mathbf{x}\cdot d\mathbf{x})^2}{\alpha^2 - \mathbf{x}^2}, \qquad \mathbf{x} = (x_1, x_2, x_3, x_4), \tag{2}$$

in which $\mathbf{x}$ is a real four-vector in the tangent space [1, Eq. (sphere3)]. Equation (2) with $-\infty \le \alpha^2 \le \infty$ describes every maximally symmetric four-dimensional geometry; the whole family is one intrinsic object read at different values of the single scale $\alpha^2 = 3/\Lambda$.

2.2 The fifth coordinate is not a coordinate of the manifold

Here is the fact the paper turns on. For every case other than the de Sitter interior ($\alpha^2 > 0$ and $\alpha^2 > \mathbf{x}^2$), the embedding coordinate $x_0 = \pm\sqrt{\alpha^2 - \mathbf{x}^2}$ is purely imaginary, and the embedding is written in an imaginary "Minkowskian" space $\mathbb{M}^5 \equiv \mathbb{C} \times \mathbb{R}^3$ by the replacement $x_0 \mapsto ix_0$. *But x_0 is not a coordinate of the four-dimensional manifold.* By (2) the manifold is coordinatised by the four real x_i, which are real regardless of the value of α^2 [1, §following Eq. (sphere3)]. The imaginary fifth coordinate is a property of one *embedding picture* of the surface, not of the surface. This is established in [1], §sec_RPT.

2.3 The Lorentzian signature is intrinsic to the positive curvature

The metric read off (2),

$$g_{ij} = \frac{1}{\alpha^2 - \mathbf{x}^2} \begin{cases} \alpha^2 - (\mathbf{x}^2 - x_i^2), & i = j \\ x_i x_j, & i \neq j \end{cases}, \qquad \lambda = \frac{\alpha^2}{\alpha^2 - \mathbf{x}^2}, \tag{3}$$

has three positive eigenvalues always, and a fourth eigenvalue λ whose sign is fixed by the real data α^2 and $\mathbf{x}^2$: $\lambda > 0$ (Riemannian) when $\alpha^2 < 0$, when $\Lambda = 0$, or when $\alpha^2 > \mathbf{x}^2 > 0$; $\lambda = 0$ (null) when $\alpha^2 = 0$; and $\lambda < 0$ (Lorentzian) exactly when $\mathbf{x}^2 > \alpha^2 > 0$. The signature is thus read from the *real* metric of the *real* manifold; no imaginary coordinate is doing the work. *And the two ways that sign can turn are of different kinds, which is worth separating.* At fixed $\alpha^2 > 0$ the Riemannian–Lorentzian crossing occurs at $\mathbf{x}^2 = \alpha^2$, where λ passes through a *pole*: $|\lambda| \to \infty$ and the sign flips, the numerator being the constant α^2 throughout. *So the metric does not degenerate at the crossing*—a signature change through $\lambda = 0$ would send $\det g$ to zero

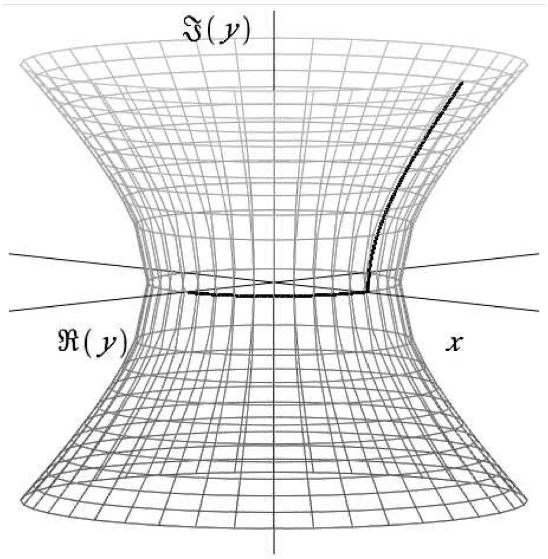

Figure 1: The maximally symmetric surface as one continuous real manifold: a spherical (Riemannian) cap joined at the equatorial throat to the Lorentzian de Sitter horn. The ninety-degree pivot at $r = \alpha$ is a coordinate event on a single real surface, not a boundary of it. (2012 dissertation, §sec_RPT.)

and leave no metric there, and that is not what happens here. The one place λ does vanish is $\alpha^2 = 0$, the null cone: not a crossing within one member of the family but the family's own singular leaf[L19]. This yields the proposition the programme's ontology stands on:

Proposition 1 (The unique intrinsically Lorentzian maximally symmetric manifold; [1] §sec_RPT)**.** *De Sitter space,* $\mathbf{x}^2 > \alpha^2 > 0$*, is the only real Riemannian manifold that is maximally symmetric and carries an intrinsic Lorentzian signature. It has a real coordinate basis, satisfies the maximal-symmetry requirement, and is the only such manifold with the requisite causal structure. This is proved here from the maximal-symmetry requirement together with the signature and causal-structure conditions; nothing is assumed of the embedding.*

The corollary is the one the dissertation flagged as a speculation and then proved: the minimum radius of the surface and its Lorentzian signature are *intrinsic properties resulting specifically from the positive curvature* [1, §near Eq. (max_symm_cont)]. The locus $r = \alpha$ where a spherical chart of (2) appears singular is a *coordinate* singularity: the curvature is finite there, the line-element "pivots ninety degrees" and continues, and what changes is the chart, not the manifold (Fig. 1). De Sitter is the one-sheeted hyperboloid with a minimum radius; the signature and the throat are what its positive curvature *is*, intrinsically, when read on the real manifold rather than imposed by hand.

2.4 The one scale, read projectively: the Cayley–Klein absolute

That one scale has a classical name, and giving it settles what kind of object α is. Read projectively, the substrate is a Cayley–Klein[L17] geometry [5, 6] whose *absolute* is the quadric $\eta(X, X) = 0$—the null cone—and its geodesic separation is exactly the Cayley–Klein log-cross-ratio taken against that absolute, with the Cayley–Klein constant equal to α: for a timelike pair, $\tau = (\alpha/2)|\log \mathrm{CR}|$; for a spacelike pair, the corresponding angle; and the discriminant $4\alpha^4(u^2 - 1)$ in $u = \eta(P, Q)/\alpha^2$ decides which, according as the joining line meets the absolute or misses it[P17R11]. *So α is not a length inserted into a metric but the constant of a projective geometry whose absolute is the light cone*, which is the same fact the signature proposition above states metrically.

A second projective fact follows at once, and it settles how the background sits in the substrate. Since a point P of the substrate has $\eta(P, P) = \alpha^2 > 0$, its polar hyperplane

$\{X : \eta(P, X) = 0\}$ carries signature $(1, 4)$, so the substrate cut by it is a totally geodesic dS_4 of isometry $SO(4, 1)$[P17R12]. That subgroup is a sweep-subgroup of the substrate's $SO(5, 1)$, which is the criterion by which a four-geometry is a *cut* [20]; and being the maximally symmetric such cut it is the dS_4 background itself. *So polarity is how the background sits in the substrate*—the background is the polar slice of a substrate point, one for each point and none privileged, the point a slice is polar to being what selects it. *It is the same pole–polar relation the throat protocol already runs on, read one dimension up.*

3 The Λ-completed vacuum as the universal standard of physics

The real manifold of §2 is not an inert stage. Its curvature radius $\alpha = \sqrt{3/\Lambda}$ is the one length the vacuum carries intrinsically, and it is against that length—and against the null cone locked to it (§5)—that everything else is measured. This is the physical content of the cosmological constant, and it is why the substrate is the object it is. The argument is Eddington's, sharpened by the century since and by this corpus's results.

3.1 Eddington's intuition, at weight

Eddington read Einstein's law $R_{\mu\nu} = \Lambda g_{\mu\nu}$ as the statement that the radius of curvature of empty space is a constant length $\sqrt{3/\Lambda}$ in every direction at every point [9]. He then made the move that matters: length is not absolute, so "constant length" can only mean constant *relative to the material standards of measurement.* Inverted, the law reads

> *The length of a specified material structure bears a constant ratio to the radius of curvature of the world at the place and in the direction in which it lies.*

So the electron's radius in any direction is a numerical constant times the radius of curvature of spacetime in that direction, and—his phrase—"an electron could never decide how large it ought to be unless there existed some length independent of itself for it to compare itself with." Hence $\Lambda \neq 0$ is not optional: without a finite intrinsic radius of curvature there is no standard against which a material structure could be what it is. The same reasoning takes the light cone $ds = 0$ as the standard of symmetry: "symmetry itself can only be relative," and the propagation of light is the standard it is relative to. This is Eddington's argument [9].

This is the intuition Eddington packed into "to drop the cosmical constant would knock the bottom out of space" [10]—a remark a later generation dismissed [11], and which Proposition 1 vindicates precisely: setting $\Lambda \leq 0$ removes the one real maximally symmetric manifold whose Lorentzian signature is intrinsic, and so knocks the relativistic metrical structure out of the vacuum.

3.2 Where the intuition was right, foggy, and wrong

Eddington reached the right object and misread its status, in ways the present construction lets us correct sharply.

Right, and now unified. He had two standards—a length standard (the radius of curvature) and a symmetry standard (the light cone)—and left them side by side. The equilateral-pseudosphere result unifies them: for the one-sheeted hyperboloid, maximal symmetry forces the equilateral profile $a = b$, whose asymptote is locked at the null cone and whose waist is the curvature radius, so c (the null-ruling gauge) and Λ (the scale) are the asymptote and waist of a *single* surface, meeting in the rate $H = c\sqrt{\Lambda/3}$ (§5; [18]). Eddington's length standard and light-cone standard are one standard, because the substrate that carries them is one maximally symmetric object.

Foggy, and now overturned. Eddington concluded that "the symmetry and homogeneity expressed by Einstein's law is not a property of the external world, but a property of the operation of measurement"—the anisotropy of a skewed world would enter both the measured structure and the measuring rod and cancel. That deflation is the one thing the corpus directly overturns. By §2 the maximally symmetric substrate is the *real intrinsic ground state* of the vacuum ([1], whose maximal-symmetry result is stated for the four-dimensional background; the void's basic state a real maximally symmetric manifold), not an artefact of comparison. The symmetry is a property of the world; and it is *because* it is a real property of the world—one standard, the same at every point—that measurement is universal. Eddington had the dependence backwards: universality of measurement is the *consequence* of a real symmetric substrate, not the *source* of an apparent one.

Wrong, and now corrected. Eddington held that there is "no radius of curvature in a timelike direction" (he was working in Euclidean coordinates with x_4 imaginary time), and drew from it that an electron, having no temporal standard to measure itself against, "just goes on existing indefinitely." The error is exactly the imaginary-time misreading §4 dissolves: de Sitter's Lorentzian signature is intrinsic (Prop. 1), its time-dimension as real as its space-dimensions, so there *is* a radius of curvature in the timelike direction. The vacuum is a consistent measuring stick throughout space *and* time [1]—and that intrinsic temporal standard is the objective cosmic-time lapse the programme later measures directly from the redshift isotropy [16]. The electron's persistence is not the absence of a temporal standard; the temporal standard is there, and it is the same maximal symmetry.

3.3 The universality of physics is the maximal symmetry

Collecting the corrections gives the physical statement the geometric core carries. The Λ-completed vacuum equation $R_{\mu\nu} = \Lambda g_{\mu\nu}$ furnishes, in its unique intrinsically Lorentzian maximally symmetric solution, the complete universal standard: the intrinsic length α, the intrinsic null structure c, and—correcting Eddington—an intrinsic temporal extent as real as the spatial one, one standard isotropic about every point. *That a specified material structure is the same structure everywhere*—an electron bound for the Coma cluster remaining an electron [1]—*is the statement that there is one standard everywhere, which is the statement that the substrate is maximally symmetric.* The universality of physics is not an added postulate laid over the geometry; it is the maximal symmetry of the substrate, read as the constancy of the standard. And its necessity is Eddington's "Λ cannot be zero" grounded: a world that exists and evolves requires a real intrinsic standard throughout space and time, which is the necessary-and-sufficient augmentation of general relativity the programme proves and measures [16, 18]. Identifying universality-of-physics with the substrate's maximal symmetry is the physical face of the paper's thesis; each ingredient (the intrinsic standard, its necessity, the measured temporal half) is established as cited; the identification is the hypothesis.

3.4 The horizon problem, properly stated

The same result settles, in the language cosmology already has a name for, a problem usually posed the wrong way round. The horizon problem is standardly stated as: why is the cosmic microwave background uniform to a part in 10^5 across regions that, on the standard expansion history, were never in causal contact? So posed, it demands a *mechanism* to communicate a common condition across a horizon, and inflation supplies one. But the deeper form of the question is not about temperature: it is why the electrons and photons at the last-scattering surface are the same electrons and photons, obeying the same physics, as those on Earth—how a causally distant patch *knows how to be* what our patch is. That is precisely the question the preceding subsection answered for an electron bound for the Coma cluster: a specified material structure is the same structure everywhere because there is one intrinsic standard everywhere, which is the statement that the substrate is maximally symmetric.

On the geometric core there is nothing to communicate. The universal standard—the curvature length α, the null cone locked to it, the real temporal extent—is an intrinsic property of the one maximally symmetric substrate (Prop. 1), present at every point by the symmetry itself, not a condition that had to propagate from region to region and could therefore fall foul of a horizon. The uniformity is not the residue of an early causal contact; it is the maximal symmetry of the substrate, the very fact that makes physics universal. CR thus does not *solve* the horizon problem by finding a faster channel; it *dissolves* it, by locating the uniformity in the intrinsic standard rather than in a shared past—there was never a gap to bridge. *The uniformity of the microwave background is read here as an instance of the substrate's maximal symmetry*, which places it at the same altitude as the universality-of-physics identification above: the intrinsic standard is established (Prop. 1), and the identification is the same reading in a second application rather than a further result.

4 Reached through the imaginary, real everywhere it lands

Proposition 1 recasts every imaginary variable in the construction as an instrument, not an ontological ingredient. Three instances, each already in the corpus, are the same move:

The embedding coordinate. $x_0 \mapsto ix_0$ makes the surface easy to see as a hyperboloid in $\mathbb{M}^5$; by §2 it adds nothing to the manifold, which is the real $(\mathbf{x})$ surface either way. The Riemannian and Lorentzian members differ by their *extrinsic* embedding curvature (Fig. 2); intrinsically each is a real manifold whose signature (3) reads off its own real metric.

The equatorial seam. Where the slicing curve runs tangent to the throat, the Riemannian cap joins the Lorentzian horn by the analytic continuation $\theta \mapsto \pi/2 + i\psi$, $\sin\theta \mapsto \cosh\psi$; the signature of the *two-dimensional slicing surface* flips because $d\theta = i\,d\psi$ squares the continuation factor to -1 [14, prop. flip]. The guard the slicing paper states, and that this paper generalises, is decisive: *the spacetime is Lorentzian throughout*; the flip is confined to the 2-surface, and the continuation is a real analytic continuation (sin and cosh one function), not a Wick rotation into a Euclidean spacetime. The seam is a real feature of the real manifold, reached through an imaginary angle.

The horizon radii, where the instrument is not optional. The three bullets above are places the corpus *chooses* an imaginary route and a reader may fairly answer "then do not take it". There is a fourth instance where that answer is unavailable, and it is the construction's own central algebraic object. Over the field $\mathbb{Q}(2M)$ of the mass parameter the horizon cubic $r^3 - r + 2M$ is irreducible—by the Gauss's-lemma argument the groupoid paper gives, which is a statement about degree in $2M$ and is indifferent to whether the base is taken real or complex [15, §galois]—and below the Nariai mass its discriminant $4 - 27(2M)^2$ is positive, so its three roots are real and distinct. That is exactly the hypothesis pair of *casus irreducibilis*, whose conclusion is that no root of such a cubic lies in any real radical extension of the base: *the three real horizon radii admit no expression in real radicals as functions of the mass*, and every radical route to them passes through $\mathbb{C}$. So for this object the imaginary is not an instrument of visualisation but the only available road, and the roots it lands on are real—which is the law of this section, met in the one case where it could not have been arranged.[9L16] *The regime dependence is the same asymmetry the groupoid paper reads off the monodromy*: above the Nariai mass the discriminant is negative, one horizon is real, and its radical expression is real again—so the forcing belongs to the under-critical side, where the family's S_3 acts on real labels, and not to the other, where only the transposition does [15, prop. deck]. *No selection is claimed from this.* The specialised cubic is reducible with real-radical roots at $M = 0$ and at the Nariai mass, but also at $2M = 3/8$ and at every rational r_0 through $2M = r_0 - r_0^3$; the reducible masses are dense in the parameter

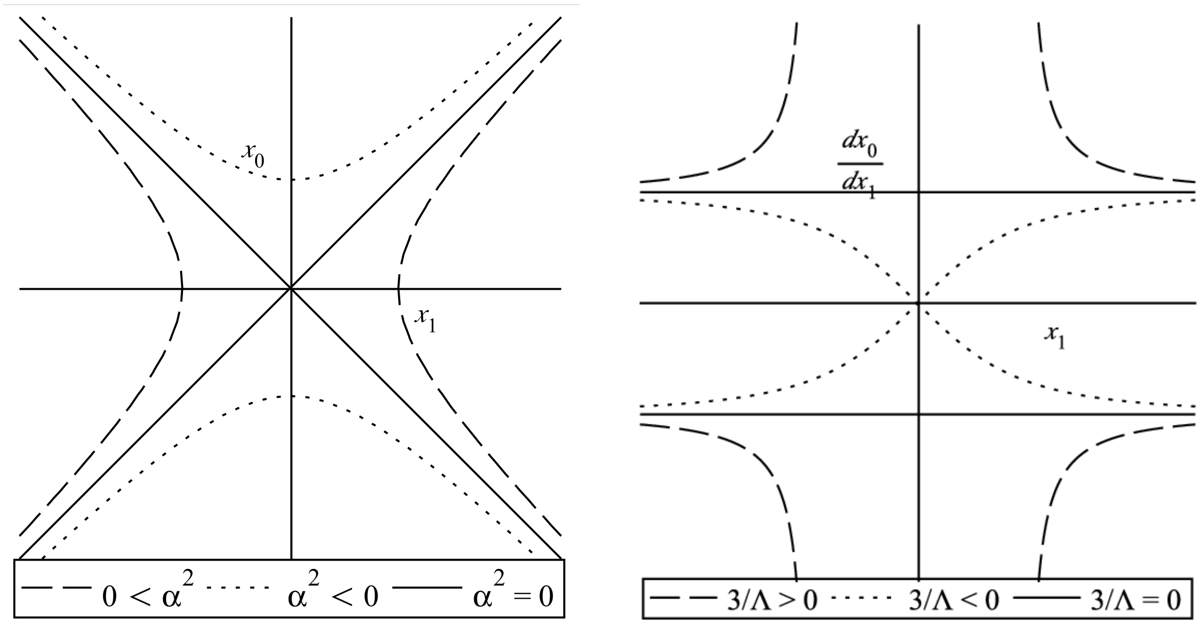

Figure 2: The α^2 family as spherical hypersurfaces of the embedding space (left) and their differential character (right): de Sitter ($\alpha^2 > 0$), anti-de Sitter ($\alpha^2 < 0$), and the null cone ($\alpha^2 = 0$). The members differ by extrinsic embedding; each is intrinsically a real four-manifold by (2). (2012 dissertation.)

and pick out nothing. The forcing is a statement about the family over $\mathbb{Q}(2M)$, where there is no specialisation to escape through.

The cosmogenesis reassignment. The correspondence that makes collapsed matter a universe is a signature-*preserving* causal reassignment on the real Lorentzian horn; the compact Wick face, where a continuous $\mathfrak{su}(3)$ would live, is real by construction but is not a co-equal existent (it carries no clock) [26, §decoupling, §face-status]. The matter rides the real horn; the imaginary face is a face, not a second reality.

Remark 2 (the general statement)**.** Each bullet is a place the corpus already refuses the unreal reading. The paper's contribution is to state the *general* law they instance: the construction reaches a real manifold through imaginary instruments, and the geometry is intrinsically real at every point they land on—substrate, seam, cosmogenesis, and the phase structure hypothesised at the seam relative to trajectories. This is CR's *built-by-construction-and-real* register; reading the imaginary parametrisation as rendering anything *unreal* is the invented flaw the discipline is built to defeat [17]. The substrate, the seam as a geometric object, and the cosmogenesis reassignment are established at the sources above, and so is the last of them: the phase structure is real as geometry and does not attach to trajectories. The reality is the plate's—R and K are the two axis-symmetries of one analytic object, and their composite supplies every kinematic datum of charge conjugation with the self-conjugate photon congruence as K's fixed set. The trajectory reading fails for a reason of its own: a massive worldline's frequency enters as m^2a^2 and the branch point annihilates it while $|aH|$ diverges, so $\omega/|aH| \to 0$ independently of m—a massive trajectory carries no phase because it fails to freeze.

5 The straight null rulings: the light cone is the visible peak

The most visible face of the intrinsically real geometry is its null structure, and it was drawn in the same dissertation chapter. On de Sitter the null condition $0 = -dx_0^2 + \sum_i dx_i^2$ makes the null-lines *straight lines* on the curved real surface in the flat embedding: a null particle traverses an angle $\pi/2$ on $\alpha < r < \infty$, and two null-lines separated by π at the throat run parallel through the embedding and never meet [1, Eqs. following (null_eom)]. The surface is *doubly ruled by straight null lines*, and the rulings are the reassigned generators the null-boundary correspondence acts on [25].

This is the geometric substrate of the c–Λ result the framework paper now carries [18, §662-region]: for a one-sheeted hyperboloid, maximal symmetry forces the *equilateral* member $a = b$—the profile whose rulings meet perpendicular at the throat and whose asymptote is locked at the null cone—so c is the null-ruling gauge, Λ the sole free scale, and the two meet only in the expansion rate $H = c\sqrt{\Lambda/3}$, with the curvature–tilt relation $\tan\theta = a\sqrt{-K}$ specialising to the locked cone. What §2 adds is the ontological reading of this: the rulings are *real straight lines on a real surface*; c fixes their slope, the throat curvature $1/\alpha^2$ their scale, and both are intrinsic to the one real equilateral pseudo-sphere. *Maximal symmetry has a dynamical reading too, and it is the strongest available.* A maximally symmetric D-manifold carries the largest isometry algebra any D-manifold admits, of dimension $D(D+1)/2$, and each Killing vector contributes one linear first integral of the geodesic flow: *fifteen integrals for the substrate's five degrees of freedom, ten for the background's four.* Liouville integrability asks for D in involution and maximal superintegrability for $2D-1$; both are met with room to spare, $15 \geq 9$ and $10 \geq 7$. *So the substrate's geodesic flow is not merely integrable but maximally superintegrable*—the dynamical face of having no distinguished point or direction, since the isometry group then acts transitively on the unit tangent bundle and all geodesics are one geodesic seen from different vantages. *Every cut lowers the count*: the isotropy strata of the algebroid paper are that descent, from fifteen here to the wall's zero, where only the norm survives [24][L14]. The c–Λ structure of the framework paper is the tip; the intrinsically real ruled geometry is the iceberg beneath it. The rulings are established in [1]; the equilateral-forcing and c–Λ [18]; the ontological reading is collected here.

5.1 The power of a point is the height: the null structure in Euclid's own quantity

The rulings are the substrate's null structure seen from outside. There is a second face of the same fact, and it is older than the geometry: the null condition on this surface *is* a classical theorem about circles, and the theorem is about the throat.

Read the substrate's overhead projection—the equatorial plane, on which the throat is a circle of radius α. For a point P of the projection at distance $|X|$ from the axis, the *power* of P with respect to that circle is

$$\mathrm{pow}(P) \;=\; |X|^2 - \alpha^2, \tag{4}$$

the invariant Steiner attached to a point in 1826 [7]: the constant value of $\overline{PA}\cdot\overline{PB}$ along every line through P meeting the circle at A, B, and, for P outside, the square of the tangent length (Euclid III.36 [8])[L1]. But the hyperboloid $-x_0^2 + |X|^2 = \alpha^2$ says that the same combination is the point's own height,

$$|X|^2 - \alpha^2 \;=\; x_0^2. \tag{5}$$

So $\mathrm{pow}(P) = x_0^2$: *the power of a point with respect to the throat is the square of its height in the embedding.* The tangent from P has length $\sqrt{\mathrm{pow}(P)} = |x_0|$; it therefore runs exactly as far across as the point stands high, and in a metric of signature $(-,+,\dots,+)$ that is $\mathrm{d}s^2 = -x_0^2 + |x_0|^2 = 0$.

Proposition 3 (The tangent is a null line)**.** [L1] *The tangent line from any point of the substrate to the throat is a null line of the embedding. Equivalently: Euclid's tangent–secant relation and the null condition are the same equation, and the only thing that turns one into the other is the minus sign in the metric.*

Proof. $\text{tangent}^2 = |X|^2 - \alpha^2$ is Euclid III.36 read through Pythagoras on the radius–tangent right angle; $x_0^2 = |X|^2 - \alpha^2$ is (5). Hence $(\text{spatial run})^2 = (\text{time drop})^2$ and $\mathrm{d}s^2 = 0$. (Verified numerically at the hinge and at 600 random heights, $\max|\mathrm{d}s^2| = 5.7 \times 10^{-14}$; receipts: `power_of_a_point.py`[P17R3], `power_is_null.py`[P17R4].) □

This is the rulings of §5 in a second register, and the two identify: a ruling meets the throat tangentially, so the rulings *are* the tangents Proposition 3 describes, and "doubly ruled by straight null lines" and "every tangent to the throat is null" are one statement. What the classical register adds is reach: the proposition is about *every* point of the overhead, not only the ones a ruling happens to pass through. The sightlines of the projection that *touch* the throat are light rays; those that *cut* it are not, and the distinction is Euclid's own—the power theorem is about secants, and the null condition is its degenerate case, where the secant's two points merge. The null character is the embedding's, and it is worth being exact about where it lives: the equatorial plane carries a positive-definite restriction of η, so no displacement *within* the plane is null. What is null is the corresponding displacement on the substrate, from the vantage lifted to its own height $X_0 = \sqrt{\text{pow}}$ to the point of tangency[P17R9]. The planar figure is the shadow of that null line, which is why the power of a point reads the light cone at all. *The tangent is exactly where the light cone is.*

Remark 4 (One privileged circle)**.** The identity $\text{pow} = x_0^2$ is not a fact about powers. It is a fact about *one* circle: the substrate's waist. A power taken with respect to any other circle of the projection is a perfectly good Euclidean quantity and is not a height, because (5) is the hyperboloid's equation and the hyperboloid has one waist. This bounds sharply what classical circle geometry can say here: a theorem whose quantity is built from the power with respect to the throat carries the signature for free; a theorem whose quantity is a chord (Ptolemy [2]), a bilinear relation among points (La Hire's reciprocity [3]), or a power with respect to some other circle (Casey's bitangents [4]) carries no height at all and returns Euclid unchanged.

The boundary is not a defect of the reading. It is the statement that the substrate has exactly one waist, met in the classical language. *For La Hire's reciprocity there is a second and sharper reason.* The hinge's polar has its foot at $\alpha^2/2\alpha = \alpha/2$, *inside* the throat, where $x_0^2 = |X|^2 - \alpha^2$ is negative and the substrate has no point at all. So the horizon-chord is real only at its ends—the two tangency points sit on the waist, at $x_0 = 0$, where the light cone touches—while its interior runs through the hole. *Only the endpoints are points of the substrate; the chord joining them is a line in the projection and not in the manifold.* La Hire's reciprocity is a statement about the whole projective plane, hole included, and its reciprocity runs through exactly the part that is not there—which is why most La Hire pairs cannot both be lifted, a point on an external vantage's polar lying generally inside the throat[L1].

The bound is not an artefact of working in the projection either. *And the classical language reaches one thing further than a power.* Euclid's trichotomy for a line against a circle—cuts, touches, misses—is the same trichotomy the slicing paper reads on its fundamental ellipse and the framework paper reads as transverse crossing, tangency at a merged double root, or no real horizon: a vertical line's intersection count with that ellipse partitions the family exactly as $\Lambda G^2 M^2/c^4$ against $1/9$ does, same middle member, and the Nariai configurations are that ellipse's vertical-tangent points [14, 18]. *One trichotomy in three vocabularies*, and the causality paper proves the tangency without a metric while the figure exhibits the equivalence with one[L1]. *What is designation-dependent is which root is called tangent, not whether the family degenerates*: the degeneration is the discriminant's vanishing and is a fact about the geometry, while the label the vantage attaches to it moves with the vantage—which is the bound that keeps the identification honest.

And the polarity has a second thing to say about the figure, which is what a projective reading is for: it dualises it.

Duality with respect to the throat carries points to lines and reverses incidence, so the pencil of slicing planes through a hinge goes over to the range of points on that hinge's polar.

That polar is computed in one line—the hinge sits at transverse 2α, so its polar is the line at $\alpha/2$—and it is not a new line in the figure: it is *the opposite side of the contact triangle*, whose vertices are the three points at which the hinge triangle's sides touch the throat, the midpoints of the companion slicing paper's midpoint-tangency statement [14][P17R7]. The dual of the hinge triangle is therefore the contact triangle, the dual of the swing is a point sliding along one of its sides, and the throat is self-dual, being the conic of the duality. The self-conjugate case is the throat-tangent door, whose pole is its own point of contact. *The corpus had the dual figure drawn from the beginning*—the tangency points are the same three the hinge placement already turns on—and read it only as a set of midpoints. The polarity of the embedding quadric itself—$P \mapsto \{X : \eta(P, X) = \alpha^2\}$, which depends on the single invariant $\eta(P, P)$ against α^2—returns at each named point exactly what (5) already gives, since that identity is the hyperboloid's equation. One further projective question closes here rather than opening, and the closure is worth stating because it is the one-scale result met from a new side. The classification of a *pencil* of quadrics by its elementary divisors asks how two quadrics sit relative to one another; applied to this figure it returns nothing, and cannot. Every conic the construction owns is manufactured from the throat—the throat itself at α, the hinge circumcircle at 2α which §5 derives as the circle on the throat's own edge through its own centre, the contact triangle's circumcircle which is the throat again, and that triangle's incircle at $\alpha/2$. Taking the throat with the hinge circumcircle and computing the pencil's degeneracies returns the doubled line at infinity, which is precisely the classical signature of concentricity[P17R8]: the classification hands back the manufacture. *A second quadric in general position would be a second scale, and the construction has one*—so the absence of a Segre problem here is the one-scale result read from the pencil side, and not a gap in the reading. The hinges are the case in point: each lies *on* the quadric ($-3\alpha^2 + 4\alpha^2 = \alpha^2$), so its polar is the tangent hyperplane there, and the tangent length $\sqrt{3}\,\alpha$ from the foot of its axis is (5) evaluated at transverse radius 2α rather than an independent relation.[P17R10]

Two ingredients of (5) are modern and it is worth marking both, since the temptation is to count one. The first is the minus sign—the Lorentzian signature, without which the tangent's length and the point's height are two numbers that happen to agree. The second is *power itself*: Euclid proves a relation among segments; it was Steiner who shifted the focus from the chords to the point and named $|d^2 - r^2|$ an invariant belonging to it [7]. Equation (5) needs the invariant, not the relation. That second move is the one this programme makes everywhere: it is the same move as reading the cut's offset r_0 not as a coordinate of the construction but as the mass itself [14, 19]. (4)–(5) and Proposition 3 are identities, verified; the one-circle bound of Remark 4 is established by the identity's own dependence on the waist; the historiographic reading is collected here. The bound of Remark 4 is not a conjecture about what classical geometry might fail to say: it was found by running the classical catalogue against the throat one theorem at a time, and the boundary is where the run stopped paying. The pole–polar relation, the radical axis of the throat and the circle that places the hinge, and inversion in the throat each return the same object—the chord of contact of a point's two null tangents, which is that point's horizon on the throat—by three routes sharing no machinery; Ptolemy, La Hire's reciprocity, and Casey's bitangent identity [2, 3, 4] each hold on the figure and return nothing, for the three reasons the remark names. The catalogue, its receipts, and the failures that drew the boundary are recorded in the programme's figure-theorem ledger ([L1], with `storyboard_receipts/`); what is carried here is the identity, its bound, and what the run returned.

Two things about that boundary are worth stating together, because they are the reason to trust the reading rather than merely to admire it. The first is that it was *found*, not posited: the classical catalogue was run against the throat one theorem at a time, and the boundary is where the run stopped paying. The second is what the run returned when it did pay. The pole–polar relation gives the chord of contact of a point's two null tangents; the radical axis of the throat and the circle that places the hinge gives the same line without mentioning a tangent; inversion in the throat gives it a third time, as the image of that circle. Three classical constructions sharing no machinery, one line—and the line is the vantage's horizon. None of the three is confined to the equatorial figure. The pole–polar relation is the quadric's own, and returns there what (5) already gives[P17R10]; inversion in the throat extends as the algebraic identity $\eta(P, P^*) = \alpha^2$ for $P^* = \alpha^2 P/\eta(P, P)$, holding off the equatorial plane and at either signature of $\eta(P, P)$, an involution whose fixed set is the whole substrate of which the throat is the equatorial section[P17R6]. The bound of Remark 4 therefore governs all three routes in the full space exactly as it governs them in the projection. Euclid's tangent–secant theorem, read on the same figure, locates the light cone: along a sightline through the throat the near intersection is timelike-separated from the vantage and the far one spacelike, and the interval vanishes at the geometric mean of the two segments—exactly the quantity III.36 identifies as the tangent length. The triple-angle identity returns the Nariai configuration. The nine-point circle of the hinge triangle *is* the throat[L1].

None of that is new physics, and that is the point. Every one of those results is already established in this corpus—the horizon, the light cone, Nariai, the throat's radius, the perspectival reading of the mass—and each is reached here by a route sharing no machinery with the one that established it. A reading that returned new physics from a two-thousand-year-old theorem about circles would be an object of suspicion; one that returns what the construction already proved, repeatedly and from unrelated directions, is doing what a faithful dictionary does. The classical geometry is therefore not evidence for the physics, and is not offered as any: it is a second way of seeing an object the construction arrived at first, and the agreement of the two readings is the only thing either can honestly be cited for. Nor is the reach of the agreement a mystery to be marvelled at. That so much of the classical theory of the circle turns out to speak about causal structure is accounted for entirely by Remark 4: it speaks about causal structure exactly when it speaks about the power of a point with respect to this one circle, and this one circle is the waist of the substrate.

6 Maximal symmetry, worn seven ways

The reality of the substrate is the reality of *the* maximally symmetric Lorentzian manifold (Prop. 1); its standard-hood is that same manifold serving as the universal measure (§3). We now state the third face: that maximal symmetry—the isometry group $SO(5, 1)$ complete and exhausted—is the single root of seven results the corpus establishes separately. And the list has a geometric substrate of its own worth naming before it is read: the four-geometries it ranges over are the polar slices of the substrate's own points (§5), one background for each point and none privileged, so *the many ways the substrate is read are indexed by the substrate itself*[P17R12]. The seven are each established; reading them as one fact is this paper's thesis, decidable by the test in §9.

1. **Necessity and sufficiency of the augmentation.** CR is the necessary and sufficient augmentation of general relativity for a description of structural existence, the necessary half measured [16, 18]. Maximal symmetry is what makes the substrate the least-arbitrary vacuum such a description can be cut from—least-arbitrariness being the programme's own criterion of necessity (Rule 2 of [17], read in the ontological register): a symmetry-breaking modulus is the adjustable parameter that criterion rejects, and maximal symmetry the unique structure that requires its configuration, so the substrate's selection is an instance of the discipline's own load-bearing rule rather than an added axiom.

2. **The constraint algebra is the symmetric-space grading.**

General relativity's own hypersurface-deformation (Dirac) algebra *is* the symmetric-space grading $\mathfrak{so}(5,1) = \mathfrak{h} \oplus \mathfrak{m}$ of the substrate—two normal deformations bracketing into a tangential one being exactly two coset directions bracketing into the isotropy, term for term—and the "wrong-sign" structure function standardly recognized at the root of the problem of time *is* the substrate's own Lorentzian coset metric, so the obstruction dissolves into a single geometric fact: the base-dependence of that metric [24]. *Named in the standard language the chain is shorter still*: for a homogeneous space the action algebroid[L5] of G on G/H *is* the Atiyah algebroid of $G \to G/H$, so the grading is the exact sequence $0 \to \mathfrak{h} \to \mathfrak{so}(5,1) \times \mathcal{C} \to T\mathcal{C} \to 0$ with $\mathfrak{h}$ the kernel of the anchor and $\mathfrak{m}$ its image—and *a splitting of that sequence is what a connection is*, so a choice of lapse and shift is a connection in the standard sense. *The problem of time is then the statement that this bundle carries no flat one*: the structure function varies over the space of cuts $\mathcal{C}$, so no splitting is integrable, and there is no canonical time because there is no canonical horizontal distribution. Maximal symmetry, read at the canonical rung, is what GR's constraint algebra already *was* and what dissolves its problem of time—established on the symmetry-reducible sector, the finite $\mathfrak{so}(5,1)$ reduction.

3. **One scale; the constants are gauges.** *The hinge figure makes the point before the constants do*: its incircle is the throat, at α; its circumcircle passes through the hinges, at 2α; its excircles stand at 3α. Those are one scale read at $1:2:3$, not three—each is α times a dimensionless constant of the equilateral triangle, so neither 2α nor 3α carries information α does not[L1]. The substrate carries a single dimensionful scale, Λ; the constants through which physics is written on it are unit gauges. c is the null-ruling slope of the equilateral pseudo-sphere (§5), meeting Λ dimensionally only in $H = c\sqrt{\Lambda/3}$; G never appears alone but only as the source's gravitational radius GM/c^2, the mass M perspectival and fixed to the Nariai value by Λ in the cosmology ($\Lambda G^2M^2/c^4 = 1/9$) [18]; $\hbar$ enters only at the seam, scaled by Λ alone, the de Sitter horizon's thermal state [29] closing the scale factor's lone self-adjoint-extension freedom *without a free parameter* [22]—and *lone* there is a dimension count rather than a manner of speaking, the radial operator having deficiency indices $(1,1)$ so that von Neumann's extension theory gives a $U(1)$ of self-adjoint extensions and exactly one real parameter, which is the parameter the horizon's thermal state spends[L10]; and k_B converts that horizon temperature to energy. So the gravitational–cosmological–quantum sector spends *no free dimensionless constant*—every would-be free parameter is a unit gauge or a freedom maximal symmetry has already spent (the perspectival mass fixed at Nariai, the quantization closed by the horizon). *The claim is about those constants and carries one scope, which is stated here rather than discovered later*: the graviton tower's mode sums are regularised, and their log-scale coefficient is computed and does *not* vanish [22]. *So the sums spend one dimensionless constant.* What is measured above — the unit gauges over the one curvature scale — stands untouched by that; what does not follow is that *nothing* in the sector spends one.

   And the count that decides this reading has been run. The sector's non-symmetric residue is one and is already counted: maximal symmetry leaves a one-dimensional quotient which *is* the scale, so no constant it fails to reach remains, and the matter sector spends none of the geometric budget[9]. *What the count leaves is not a constant but an initial condition*—the measured ρ_r/ρ_m, which the corpus carries as empirical at no cost to the dissolution, exactly as flat ΛCDM carries η. The source facts are each established and cited, the sector's free-data count of zero is carried by that count, and the theory's remaining free datum is the matter's.

4. **The cosmology is parameter-free.** With Λ the sole scale, the low-multipole deficit, the transmission character, and the geometric expansion rate carry no tunable freedom—the rigidity whose two-sidedness (match on the rate and the abundances; a parameter-

free, falsifiable prediction on the acoustic shape—the low-multipole a cosmic-variance-limited wash, the damping scale a genuine open build) is the discriminating power, the one measured initial condition ρ_r/ρ_m an η-analogue *read* not tuned [25].

5. **The continuous matter symmetry is excluded.** $\mathfrak{su}(3) \not\subset \mathfrak{so}(5,1)$ precisely because $SO(5,1)$ is exhausted; the same completeness that locks the constants leaves no room in the continuous isometry for a geometric colour [26].

   The complement is not empty, and the same completeness fixes it: the substrate $\mathrm{dS}_5 = SO(5,1)/SO(4,1)$ is topologically $\mathbb{R} \times S^4$, hence spin with a *unique* spin structure, inherited by the four-dimensional cut; and the one orientation parity $R \in \mathrm{O}(5,1) \setminus \mathrm{SO}_0$ [24] reflects the cut-normal (r_0) direction, acting on the cut's natural spinor as the chirality operator γ^5 itself—the cut-normal being the transverse mass-modulus (the offset r_0 that *is* the mass), whose $\mathrm{Cl}(1,4)$ generator is γ^5—so the substrate's single orientation $\mathbb{Z}_2$ *is* the chirality grading, measuring left against right as it carries the graviton's helicities. *And the discrete residue is larger than the orientation parity alone, by a route that uses only what the substrate already carries.*

   The horizon roots carry surface gravities, hence a residue pairing on the triple; that pairing has a holonomy about the Nariai points—the per-root resolution of a square root the Galois statement takes in the product—and adjoined to the monodromy symmetry it closes the Weyl group of $\mathfrak{so}(6,\mathbb{C})$, this substrate's own complexified isometry algebra [15, 24]. *The A_2 the corpus reads off the cubic is the sub-root-system obtained by deleting one node of that A_3*, so the discrete structure is not an addition to the object but a fuller reading of it—which is this paper's thesis applied to itself.

   So the exhausted isometry does not merely wall a continuous matter symmetry; it fixes a discrete one, the fermion chirality identified with the substrate's own orientation parity. What the residue supplies is the fermion sector's arena, its chirality grading, and its discrete generation-structure—built and forced within CR rather than conjectured; the content it grades—the mass values, the electroweak scale—is the ordinary route, not the geometry's, so CR's claim is a gravitational–cosmological unification, not a geometric unification of matter [26].

   The residue's discrete content is moreover a linear orientation skeleton—the parity P, a time reflection T, the γ^5 grading, and the R-parity carrying the fundamental $\mathbf{3}$ to its antifundamental $\bar{\mathbf{3}}$—whereas charge conjugation, antilinear where those act linearly, is field-level like the charge it acts on (the substrate carrying it as a symmetry and not a datum—the metric even in the charge, its sign the bend of the cut [20]); so a geometric CPT is not auto-yielded, the geometry supplying the orientation *skeleton* of matter *and its antimatter*—the discrete $\mathbf{3} \oplus \bar{\mathbf{3}}$ the R-parity relates, matter and antimatter at one weight within CR [28]—and not the charge, which is field-level on both branches alike—the same partition once more, the residue the arena, the grading, and the skeleton, the content and the charge the ordinary route [26].

   The antilinear structure is not, however, exhausted by the field level, and the boundary the residue draws encloses a positive shape: the reality involution $\tilde{\tau} \mapsto \bar{\tilde{\tau}}$ on complexified cosmic time is antilinear *and* geometric, so charge conjugation factorises into a geometric kinematic (Feynman–Stückelberg) face—carried by the bead's own $r = 0$ crossing, so that the cosmogenesis carries C's kinematic content—and the field-level charge sign alone, a geometric CPT staying unyielded because only the charge closes from the field, and this face realised on the built fermion sector's zero-modes [26, 28]. The spin structure and the γ^5-action are grounded: the parity *measures* chirality (the exchange reading excluded by explicit computation, since R reflects the transverse cut-normal and fixes the four spacetime legs), detailed in the boundary paper [26]; the same parity grades mass in *both* faces—the geometric mass-reflection $r_0 \mapsto -r_0$ (whence $2M \mapsto -2M$) and the R-odd fermion mass term—so mass, geometric or fermionic, is the substrate's one R-odd datum. *And that identification carries further than it has been read, so we develop it here rather than leave it in a clause.* Read at the electroweak sector it says that what the *Higgs mechanism* breaks is this parity: the R-symmetric sector is the offset-free, massless configuration and mass is the R-odd departure from it. *What is thereby claimed is a mechanism and not a magnitude*—the vacuum expectation value, the electroweak scale and the individual masses stay the ordinary route, as this paper says throughout and as the one-constant structure requires. *But three things follow that were not being said, and each is a property of the slicing relation the companion paper already carries*[P17R15], $2M = r_0 - r_0^3$ in the gauge $\alpha = 1$ [14]. *First, the mass-reflection above is an identity of that relation rather than a further assumption*: the cubic is odd, so $r_0 \mapsto -r_0$ sends $2M \mapsto -2M$ exactly. *Second, the symmetric sector is the bare substrate.* $2M = 0$ at $r_0 = 0, \pm\alpha$, and the metric function is $1 - r^2/\alpha^2$ at all three—one massless geometry read at three offsets, so the unbroken phase of this breaking is de Sitter itself and not a separate configuration. *Third, and this is the part that distinguishes the geometric statement from the field-theoretic one it is being set beside: the order parameter is bounded, and its bound is the degenerate member.* Over the admissible offsets $|r_0| \leq 2\alpha/\sqrt{3}$—the regime range the slicing paper reads off the discriminant $4 - 3r_0^2$—the mass satisfies $|M| \leq \alpha/3\sqrt{3}$, with equality at $r_0 = \pm\alpha/\sqrt{3}$ and at the range endpoints, and that value is the Nariai mass, obtained independently as the double root of the horizon cubic. *A quartic potential is unbounded in its field; this order parameter is not, and it saturates exactly at the configuration a collapse reaches* [18]. *And the breaking has a cause here rather than a chosen minimum.* The companion paper does not posit the offset: it derives that a manifold observer cannot be assumed to sit at $r_0 = 0$, because the reference such an observer actually has is the hole's image on their celestial sphere and there is no marked point on that image for a reticle to land on [14]. *So on this reading the symmetry is broken because of what a reference is, not because a potential was written with its minimum away from the origin. We state the limits of that with the same care.* No vacuum expectation value, no scale and no mass value follows from any of it; nothing here derives the Higgs field, its potential, or the gauge group it breaks—and if that group were ever promoted from described to forced, the relation this paper draws to it would have to be re-examined rather than inherited [26]. One candidate centrepiece stands here: the residue's full automorphism $\mathrm{Aut}(A_2) = S_3 \times \mathbb{Z}_2 \cong D_6$, read on a fermion sector, carries the shape 3×2—three generations times two chiralities—the $\mathbb{Z}_2$ the γ^5 grading just fixed, the S_3 permuting the three mass-tied roots of the offset-mass cubic (the discrete $\mathfrak{su}(3)$-Cartan–Weyl shadow [24]), a $2M$-degenerate flavour triplet whose hierarchy would be a separate breaking; the chirality factor grounded, and the generation factor since *built* and forced within CR [28]; and the descent onto a *propagating* spinor sector is now built as well, a Dirac field on the unpolarised radiating member propagating on the light cone and carrying the twist on γ^5 alone [23]. The same A_2 skeleton is now realised in the geometry itself: the slicing paper draws the hexad as a skew hexagon of null rulings across the substrate's two horns, each horn's three vantage-hinges designating the three roots [14]. Whether that geometric six-fold is the *same* hexad as the flavour triplet here—the one exchanged by the geometric time reflection, the other by the offset parity R, two distinct $\mathrm{O}(5,1)$ reflections that both swap the null rulings—or a resonance of shared abstract type, is *settled* rather than open: the slicing paper establishes it as a resonance and *not* an identity—the horn-swap T (timelike, horn-flipping) and the offset parity R (spacelike, horn-preserving) being distinct reflections that share the fundamental triple and complete it distinctly [14], verified on the six points. The chiral content this parity would grade is not built here.

6. **One circle: the equator is what the construction is built on.**

   The substrate has exactly one waist (Rem. 4), and its radius *is* the curvature: $\alpha = \sqrt{3/\Lambda}$, the sole scale (§3).

   What the construction is built from is a set of facts about that one circle, and they are one fact.

   The power of a point with respect to it is the square of the point's height, (5), so the tangent to it from any point of the substrate is null (Prop. 3) and the rulings *are* those tangents (§5): the double ruling and the classical power law are the same statement, set by α and by nothing else, and the power vanishes identically on the circle itself ($x_0 = 0$).

   That circle is the incircle of the hinge triangle and its nine-point circle besides[L1], so the hinges stand at 2α as an *output* of the hole rather than a stipulation, the three sides are the Nariai double null rulings tangent at their midpoints, and the triple-angle identity returns the Nariai configuration of its own accord [14].

   The signed areal radius passes through zero at a point *on* that circle, so the three hinges' walls lie on it as a $\mathbb{Z}_3$-orbit whose three-foldness is the hole's own, and the fermion sector's three chiral generations are those three walls [28]. The slicing's lap runs around it, and through it the cosmological gravitational collapse of a black hole in a prior universe completes into the formation of a universe like our own—collapsed matter cannot terminate but continues as a cosmology—matter and antimatter the two ends of one conjugation across the crossing it carries [18, 26]. And the *count* is not a fifth thing read off that circle but a statement about the circle's own two relations. The substrate's discrete residue is the direct product $\mathrm{Aut}(A_2) = S_3 \times \mathbb{Z}_2$, whose two factors act on independent structures—the Weyl S_3 on the three roots, the inversion on the two rulings [24]. Read on the waist, those structures are the two things a figure can be to a circle: the three roots are the three special points *on* it (the offset line meets it at A and B, with $r = 0$ fixed at the back) [14], and the two rulings are the lines *tangent* to it (§5). So the residue factorises because *on* and *tangent* are independent[L6], and the factors differ in kind for the same reason: a ruling is a line of the substrate, so exchanging the two is a motion of it—an isometry, acting on the cut spinor as γ^5, and chirality descends *gauged*; a root labels a different cut, so permuting the three moves through the solution space and is no motion of the substrate—the monodromy symmetry *of that cover*, which is the Weyl S_3 and not the turnaround's $\mathbb{Z}_3$ the generations bear (the two threes distinguished below)—and a family symmetry is *global*. The Standard Model's arrangement of a gauged chirality against a global flavour is, in this reading, the difference between being *on* the one circle and being *tangent* to it.

   Maximal symmetry is why there is one such circle and not a family of them, and why α is the one length the exhausted isometry leaves to set it: the lines, the curvature, the placement, the count, and the passage are all read off the same waist.

7. **The universality of physics.** One maximally symmetric substrate is one standard the same at every point; that a material structure is the same structure everywhere is that constancy of the standard (§3). Universality is not a postulate over the geometry but the maximal symmetry read as the invariance of the measure [1, 16].

A consequence of the fifth face is drawn, and it is worth stating here precisely because it is *not* an eighth face. The residue's mass-parity—$r_0 \mapsto -r_0$, whence $2M \mapsto -2M$, the substrate's one R-odd datum—and the triple angle that fixes the discrete generation index are both statements about the *cut*, and written in a general dimension they cease to hold. The metric function in D dimensions is not assumed but obtained: the slicing operator's vacuum condition, which is the kernel of its matter functional on a cut in the construction gauge, reads

$rf' + (D-3)(f-1) + 2\Lambda r^2/(D-2) = 0$ in D dimensions, and its entire solution space is the Tangherlini–de Sitter family $f = 1 - 2M/r^{D-3} - r^2/\alpha^2$ with $\alpha = \sqrt{(D-1)(D-2)/2\Lambda}$ and M the single constant of integration [14]. With that function the mass is $2M = r_0^{D-3} - r_0^{D-1}$; the horizon relation collapses to a single multiple-angle in the sky angle only at $D = 4$ and $D = 5$, since the harmonics standing below the top one number two or more from six dimensions upward against a single available scale; and $2M$ is odd in the signed offset only when D is even, so at $D = 5$ the parity fixes each geometry instead of exchanging it with its conjugate and there is no R for the residue to be. Four dimensions is therefore the only one carrying both a generation count and a chirality, and the corpus's three is its $D - 1$ [14, 28]. *The reason this is not a face of maximal symmetry is that it does not come from maximal symmetry.* The substrate is maximally symmetric and moduli-free in every dimension, so the criterion of necessity has nothing to grip on the dimension itself; what selects here is the *matter sector's own content*, and the two statements are compatible only because they are about different objects—the cut's dimension is settled, the substrate's remains bounded below and not above. Reading this as a ceiling on the substrate would re-make exactly the error the distinction is drawn to prevent.

So the property that makes CR *decisive against the world*—no knob at any rung—is the property that makes its matter sector *geometrically hard*: maximal symmetry, having spent everything, leaves the cosmology nothing to tune and leaves the geometry no continuous structure to build matter from. The rigidity and the wall are one fact—and the count of free data makes it exact: with the gravitational–cosmological–quantum sector spending no free dimensionless constant *in its geometric ledger*—the graviton tower's mode sums spend one, computed at the frontier item [22], which is a regularisation and not a gauge—the theory's entire free-data budget—the one measured ρ_r/ρ_m and the fermion sector's own content—is carried by the matter, so matter is the residue maximal symmetry leaves not in field content alone but in free data. The descent onto a spinor sector [28] makes this precise at the discrete rung: the fermion sector's discrete structure—the generation multiplicity, the chirality, and the family symmetry, the full D_6 residue—is forced within CR, drawn from the substrate rather than tuned, so it leaves the free-data budget for the residue; the budget then carries the fermion *content* alone—the gauge assignment and the mass values, the ordinary route, whose deeper structure is the open search. The discrete half of matter is thus the stone's and the content half the search. The residue the wall leaves—now developed from the discrete orientation parity into that full skeleton where matter's one geometric opening lives—is taken up in §7.

6.1 The geometric content of the ledger, and the fine-tuning problems the one scale dissolves

The constant rung is not bare unit-counting. On this fully determined geometry each gauge is a specific geometric feature: c is the null-ruling slope of the equilateral hyperboloid (§5), Λ the waist, and G the cut's *offset*—the mass *is* the offset of the section from the central geodesic, $2M = \alpha\big((r_0/\alpha) - (r_0/\alpha)^3\big)$ [19, 14], so G enters only as the offset-length GM/c^2 and is the identification of the mass-label with that offset, not a coupling of independent matter. These three—c, Λ, G—are the gauges of the *real* Lorentzian geometry, each a nameable feature of the substrate and its cuts. $\hbar$ and k_B enter a different register: the de Sitter horizon's standard Gibbons–Hawking thermal state (period $\beta = 2\pi\alpha$) [22], a Euclidean continuation distinct in kind from the real-analytic imaginaries of §4. So whether "the constants are gauges" is mere unit-counting has a structured answer: the real-geometric gauges c, Λ, G carry CR-specific geometric identifications; the thermal gauges $\hbar, k_B$ are standard horizon thermodynamics, their CR-specific content the ledger's closing of the one quantum freedom, not a new geometric feature. The identifications and the register split are established at the cited sources; reading them as one partition is this paper's consolidation of them.

This settles what the Planck values *are* here. The traditional Planck length, mass, and time—$\ell_P = \sqrt{\hbar G/c^3}$, $m_P = \sqrt{\hbar c/G}$, $t_P = \sqrt{\hbar G/c^5}$—are combinations of these gauges, and *cross-register* ones, mixing the thermal $\hbar$ with the real-geometric c and G. Since the ledger leaves exactly one physical scale (Λ, equivalently $\alpha = \sqrt{3/\Lambda}$), a Planck value is *not* a physical scale but what unit-counting yields when it has only gauges to count. The one physical length is α, not ℓ_P; their ratio $\alpha/\ell_P \sim 10^{61}$ (from $\Lambda\ell_P^2 \approx 3 \times 10^{-122}$ [25]) is the size of the universe in gauge-units—a number, not a tuning.

One thermodynamic quantity the corpus has never taken belongs here, because its number is this one and a reader arriving from horizon thermodynamics will compute it. The de Sitter horizon whose Gibbons–Hawking state supplies $\hbar$ above has area $A = 4\pi\alpha^2$, so the Bekenstein–Hawking value carried on it is

$$S = \frac{A}{4\ell_P^2} = \pi\left(\frac{\alpha}{\ell_P}\right)^2 = \frac{3\pi}{\Lambda\ell_P^2} \approx 3 \times 10^{122}, \tag{6}$$

which is the gauge-count just stated, squared, and nothing further[P17R17]. *So the horizon thermodynamics introduces no scale the ledger has not already entered*: S is a pure number built from α/ℓ_P, and ℓ_P enters it as the gauge it was shown to be rather than as a physical length the entropy is measured against.

And it is not merely of the same magnitude as the number the next paragraph meets; it is the same number. The cosmological-constant problem's factor is the ratio of the field-theoretic vacuum estimate to the observed $\rho_\Lambda = \Lambda/8\pi G$, which is $8\pi/(\Lambda\ell_P^2)$; the entropy is $3\pi/(\Lambda\ell_P^2)$. *The two differ by* $3/8$ *and by nothing else*—both are $1/(\Lambda\ell_P^2)$ read with a different numerical coefficient, and the 10^{122} of the one is the 10^{122} of the other rather than a coincidence of size[9L21]. So the reader who arrives with the entropy and the reader who arrives with the fine-tuning are holding one quantity, and the dissolution given below for the second is already the dissolution of the first: there is no physical scale for the ratio to be a tuning *between*, ℓ_P being a cross-register gauge-combination and not a length the world is built to.

This also says why the corpus takes this horizon's temperature and never its entropy, which is a structural difference rather than an oversight. The temperature $T = 1/2\pi\alpha$ is built from α alone—one register, the real-geometric one, with the thermal gauge setting only the unit; the entropy is a ratio of α to ℓ_P and is therefore a count taken *across* the register split above, mixing the thermal gauge with the real-geometric ones. *The quantity the framework needs is the one-register one, and the quantity that would test the ledger is the cross-register one*—and what the cross-register one returns is the ledger's own number back. *What is not claimed is that a de Sitter entropy is asserted here*: whether $S = A/4$ carries to a cosmological horizon on this reading is a question this paper does not settle, and the clause states what the number *is* if the standard expression is taken. *Were it to fail to carry, that would be a result and not a gap*—a one-scale ledger forbidding a thermodynamic relation rather than merely accommodating one. The ledger, the register split and the Planck-value reframe are established above, and $S = \pi(\alpha/\ell_P)^2 = 3\pi/(\Lambda\ell_P^2)$ together with its coincidence with the cosmological-constant factor is computed [P17R17]. Whether $S = A/4$ applies to this horizon is not claimed.

One consequence of that declination belongs with it, because it makes the branch consequential where this paragraph leaves it costless. A topological term—the Gauss–Bonnet combination—contributes no field equation in four dimensions [22], so a coefficient multiplying one is not read off any dynamics; its one remaining observable is a horizon-entropy contribution, and that contribution is *topological*, going as the Euler characteristic of the horizon cross-section [21] and so contributing an area-independent constant. *And the ledger has no absorber for a constant entropy shift* as it has the one curvature for a constant vacuum energy: $S = \pi(\alpha/\ell_P)^2$ is fixed once α is. *So the two branches do not cost the same.* If $S = A/4$ carries, the sector acquires a free dimensionless constant that cannot be absorbed—against a ledger stated to spend none; if it does not, the coefficient has no home and the failure is the result this paragraph already anticipates. *The one-scale reading therefore has a stake in the branch, and the question is not neutral*[P17R1].

This positions the framework to make a striking conjecture, stated here as the hypothesis it is, to be grounded through the matter sector: *the one scale dissolves the two deepest fine-tuning problems of cosmology at once.* Two results of the projective bake sharpen what "one scale" means here, and they belong with the claim rather than after it. **First**, the substrate's metric is not merely *compatible* with a single scale: read projectively it *is* the Cayley–Klein construction on the absolute $\eta(X, X) = 0$, whose scale constant is α (§5)—and a Cayley–Klein geometry carries the scale as its *only* free constant, signature, null structure and isometry group all being fixed by the absolute's projective type.

The one-scale reading therefore has a classical mechanism and is not only the outcome of an exhaustive audit. **Second**, α is precisely what the causal structure does *not* fix: the group preserving the absolute is $\mathrm{O}(5, 1)\times\mathbb{R}^+$, one generator larger than the substrate's isometry group, and that extra generator is the dilation carrying $\alpha \mapsto \lambda\alpha$ [15].

So the null cone determines the geometry up to the scale, and the scale is the one thing it leaves—which is why there is a single dimensionful input and nothing for a dimensionless tuning to hide in. *And the family that dilation sweeps out has a shape worth naming, because it makes the light cone structural rather than a limiting case.*[P17R2L17L8]

The substrates of every α are the level sets of one quadratic form on the ambient, so the family is a *foliation* and the null cone is its singular member—the leaf at $\alpha = 0$—rather than an object of a different kind sitting beside the family. *One expectation about that family fails, and its failure is structural rather than a gap.* A confocal family of quadrics carries a classical orthogonality theorem, and in the equilateral case—which is what this substrate is—the confocal family *is* the dilation family, so the theorem looks applicable.

It is vacuous here: the theorem is not that confocal quadrics meet orthogonally, but that through a generic point pass *three* members, one of each type, and that those meet pairwise orthogonally. The confocal equation is cubic in its parameter generically and *linear* in the equilateral case, so exactly one member passes through any point and there is no second for it to be orthogonal to.

The hypothesis fails, not the conclusion—and the same fact without the machinery is that a point assigns one value to the quadratic form, which is what makes the family a foliation. *And that same one-member-per-point fact fixes where this construction's harmonic analysis can live*: a confocal family through which three surfaces pass at each point supports separation in ellipsoidal coordinates, and one through which a single surface passes does not. There are therefore no ellipsoidal harmonics to be had on the substrate, and the corpus's harmonic analysis is leafwise throughout—on the spatial leaf, in its own spherical harmonics—not by choice of method but because the ambient offers no other separation[L12].

The *cosmological-constant problem*—that Λ is some 10^{122} times smaller than the quantum-field vacuum estimate $\sim M_{pl}^4$—rests on two premises the framework rejects.

First, that the Planck scale is a physical scale Λ must be small against: it is a gauge-combination (above), so there is nothing physical to be small against.

Second, that a bare Λ and the matter vacuum energy are distinct quantities whose sum must be finely cancelled to the observed value. Here Λ is the geometrically primary substrate curvature—the maximally symmetric ground state's own scale, the single scale of the ledger—and a *constant* vacuum energy is not a source held against a bare Λ but is absorbed into that one observed curvature: a constant density gravitates as a curvature scale, so it enters the profile's $\Lambda r^2/3$ term, not as a $2m/r$ bend, and there is no bare-Λ-versus-vacuum-energy split for the 10^{122} cancellation to act on—the substrate carries only the total, the observed ground-state scale. The substrate/bend distinction separates Λ from *inhomogeneous* matter, a genuine bend that breaks the maximal symmetry, not from the maximally symmetric vacuum energy the ground-state curvature already carries. The first premise is defused on established ground; the second is defused on the profile structure—and it must be, since reading vacuum energy as a $2m/r$-type bend overreaches for precisely the constant vacuum energy the problem concerns. The *coincidence problem*—why Λ is comparable to the matter density now—is already answered: the framework has one timescale, $\sim 1/(\sqrt{\Lambda}\, c)$, and any observer observes at a time of its order [18]. That both of the deepest fine-tuning problems fall to the *same* one-scale fact—the substrate's maximal symmetry read at the constant rung—is a strong suggestion of coherence at the most fundamental level, and it is where the geometric core ceases to be an accounting of the corpus and becomes a place the framework renders its own verdict. The coincidence dissolution is argued in the framework paper [18] and the Planck-scale reframe above; the cosmological-constant dissolution—that there is no bare-Λ-versus-vacuum-energy split for the cancellation to act on—rests on the profile structure and on Λ being geometrically primary.

The register split is the substrate's two real forms, and this is where the ledger opens onto the physics. The real-geometric gauges c, Λ, G are features of the *Lorentzian* real form $\mathrm{dS}_5 = SO(5,1)/SO(4,1)$: the existent, temporal geometry whose cuts are the solutions of general relativity, whose indefinite signature is the "wrong sign" the constraint algebra carries as the problem of time [24], and on whose cuts the one discrete residue is read once as that solution space and again, on a fermion sector, as the flavour skeleton [28].

The thermal gauges $\hbar, k_B$ enter through the horizon's Gibbons–Hawking Euclidean, and that continuation is the substrate's *other* real form—the global Wick $x_0 \mapsto ix_0$ carrying dS_5 to the compact $S^5 = SO(6)/SO(5)$, the same compact face of §4 on which a continuous $\mathfrak{su}(3)$ can live, real by construction but carrying no clock [26].

These are two of the real forms of the one complex group $SO(6,\mathbb{C})$, meeting at the horizon where $\beta = 2\pi\alpha$.

So the thermal and the gauge registers are not two entries in the ledger but one tenant of one real form: colour requires the *full* $SO(6)$—the smallest faithful real representation of $\mathfrak{su}(3)$ is $\mathbb{R}^6$, so it cannot sit in the $SO(5)$ of the seam continuation—and the horizon's $\beta = 2\pi\alpha$ Euclidean rides that same S^5, so the quantum of action and the colour symmetry share the one Euclidean real form exactly as c, Λ, G share the Lorentzian one.

Read this way the ledger's own partition is the corpus's deepest divide seen at the constant rung. The Lorentzian real form is the existent world: general relativity, the cosmology whose fluctuations are inherited classical content rather than a substrate quantum vacuum [25], the matter, and the flavour the residue grades. The Euclidean real form is the atemporal structure that world carries—the gauge group and the quantum scale together—real but not a co-equal existent, because it holds no clock (§4; [22]). The maximal symmetry this section has worn seven ways is here worn once more and most simply, as the pair of real forms itself: one substrate, one complex group, the temporal world and its timeless structure its two real slices. On this reading the two divides the last century drew—general relativity against the quantum, and gravity against the gauge forces—are not theories to be joined but the one substrate read on its two real forms, the join already present in the complexification the substrate was reached through; the physics of that reading is drawn in the boundary paper, now the corpus's physics-synthesis [26], and its geometric closure is this. The register split and the two real forms are established (§4; [26, 24, 22]), and the co-location of the thermal and gauge Euclideans—$\mathfrak{su}(3) \subset \mathfrak{so}(6)$ but $\not\subset \mathfrak{so}(5)$—is computed[P17R5].

7 Physics as broken-symmetry shadows

If the maximally symmetric substrate is the universal standard (§3), then the physics—the exact geometries, the masses, the warping, the cosmic history—is what that standard looks like when its symmetry is cut. This is the frame the paper is written in, and it is already the corpus's, stated here in one place.

The ladder of symmetry. The substrate is $\mathrm{dS}_5 = SO(5,1)/SO(4,1)$, maximal symmetry complete. Its physical leaves carry the maximally symmetric four-dimensional background dS_4—

the empty cosmology, the clock running on Λ alone.

Below these are the exact solutions: Schwarzschild, Schwarzschild–de Sitter, Nariai, the Friedmann geometries, each obtained by a cut that breaks some of the substrate's symmetry, with matter the bend of the cut that no continuous symmetry holds in place [14, 19, 20]. Each descent down the ladder is a symmetry broken; the standard above is exact, the geometry below is the standard read through the break. *What kind of relation the word "cut" names is worth fixing here rather than leaving to be inferred, because the two available readings part company one rung down and the parting is computable.*[P17R14] The first descent is *linear*: a hyperplane section $\{X : \eta(n, X) = c\}$ of the substrate returns, on splitting $X = cn + Y$ with $\eta(n, Y) = 0$, the locus $\eta(Y, Y) = \alpha^2 - c^2$—a de Sitter four-space of radius $\sqrt{\alpha^2 - c^2}$, for *every* admissible n and c, a quadric cut by a linear space being a quadric.

The polar hyperplane above is the $c = 0$ member, and the computation shows it is not one example among many but the *only* kind: since Schwarzschild–de Sitter has Kretschmann scalar $48M^2/r^6 + 24/\alpha^4$, which is constant only at $M = 0$, *a plane section of the substrate is Schwarzschild–de Sitter exactly when the mass vanishes. Below that rung the relation is therefore not a section, and it is not an isometric embedding either.* Imposing on a would-be hypersurface only what the slicing construction already carries—a round areal two-sphere, and staticity as the ambient boost the pencil turns on—makes the substrate condition fix one radial function and the induced metric fix its derivative, two determinations of one function; they disagree, and the boost rate they would require is radius-dependent, which a Killing flow's rate cannot be. *The closure defect factors with an overall factor of M*: the amount by which a cut fails to close as a hypersurface is the mass it carries, which is this framework's "matter is the bend of the cut" as an identity rather than a reading. This is consistent with the classical embedding class—Schwarzschild needs two flat dimensions above four and the substrate supplies one—and it fixes the usage: *"cut" is the group-theoretic relation the companion papers define*, a geometry whose isometry group contains a sweep-subgroup of the substrate's [20], fixed by an orientation datum together with a causal-vantage datum [24], with the derived metric read through that break. *The isometric-embedding reading coincides with it at $M = 0$ and nowhere else*, and nothing in the programme rests on the stronger reading. Among those cuts are both the black holes and the cosmologies, so a black hole and a cosmos are not two objects but one—the substrate read through two breaks—and the collapse of one is continued through the seam as the expansion of the other, the *cosmogenetic bead* the framework proves one closed curve of the substrate [18]. The corpus's deepest single identity is at bottom a fact about the one object: collapse and cosmology are the same substrate seen twice.

The cosmological rate rule is this frame, computed. When the cosmology carries out a calculation on this ladder it reads three things off the one object, and they are the shadow-reading made operational. The *layer* is the existent—its rate of advance, the lapse, the foliation stacking rate set by the geometry alone, because the standard is exact and the content is what the set rate carries, not a term that sources it. The *projection* is the shadow the observer reads—the synchronous appearance through which that advance is seen as distance and redshift, a chart and not the object. The *bend* is the matter, the leaf's departure from the round standard. So the three-level rule the cosmological papers compute on—the observable expansion riding the geometry-set rate, the plasma's diffusion and sound horizon carried on the layer and then projected, never the shadow's machinery run as the existent—is not a separate device but this section's frame at the level of a rate [16, 22, 25, 27]. The rigidity that makes the standard exact is what leaves the cosmology no knob (§6); the same exhaustion of continuous symmetry that walls the matter sector is what fixes the rate from the geometry, so the geometric rate and the geometrically hard matter sector are the one maximal symmetry read at two rungs.

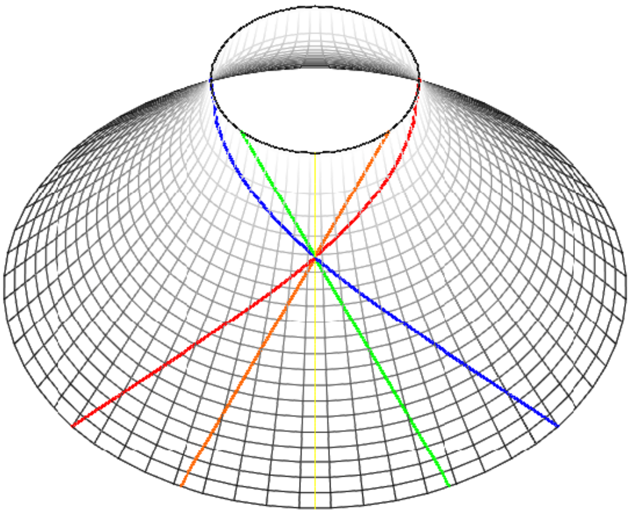

Figure 3: A cut of the substrate read as a universe: comoving worldlines and their null structure on the real de Sitter surface. The exact cosmology is a symmetry-breaking slicing of the maximally symmetric object, not a separate geometry. (2012 dissertation.)

Warping as perception — the shadow off the wall. The dissertation reached the reading this makes precise: the possibility that the void does not dynamically warp in the presence of mass at all, but is de Sitter space with the universe moving uniformly through it, the apparent warping of spacetime around massive objects being "the perception of things therein," by the principle of equivalence [1]. The corpus makes this a discipline rather than a suggestion: the exact geometry a given observer charts is a *projection* of one intrinsically real substrate, the perspectival appearances read off the wall by the shadow-reading epistemology, with what is literal and what is perspectival separated exactly [17, 18]. Schwarzschild mass is the cleanest instance: it is the perspectival label a particular swept vantage manufactures on the fixed-α manifold, real by construction and observer-dependent, not a property the substrate carries in itself [14, 15]. The warping is a true shadow—built by construction and really seen—of a real, unwarped, maximally symmetric object.

Matter's one geometric home. The break has a residue. Maximal symmetry, exhausting the continuous isometry, leaves over exactly the *discrete* orientation parity $O(5,1) \setminus SO_0(5,1)$— the $\mathbb{Z}_2$ that threads the mass-reflection, the graviton helicities, and the chirality of the turning polarisation plane past the wall [15, 23, 24, 26]. That discrete residue is the structurally indicated home of a fermion sector, if the geometry carries one at all: not the continuous isometry the wall closes, but the discrete parity the wall cannot reach. This is not claimed in either direction, and the ordinary route — a matter bundle placed by hand — is untouched by the wall.

The generation conjecture, and the within-CR result it became. One reach on that residue is worth stating whole, as the conjecture it is, because its parts are grounded in several registers at once and only the last step is open—and capturing it coherently is how the corpus takes hold of a broad, tenuous structure to study it. **The reading, and what stands behind it.** *The Standard Model's three fermion generations are the substrate's own three-fold slicing structure, read on a spinor sector*—the three mass-tied roots of the one R-odd offset-mass cubic (the A_2 weights), equivalently the three 120°-separated hinged vantages of the one sliced substrate, carrying the chirality $\gamma^5 = R$. What is *grounded*, in three registers:

- *Geometry, the multiplicity.* The count is forced to three: the offset-mass relation is the pure triple-angle $2M = \frac{2}{3\sqrt{3}} \sin 3w$, driven to a single harmonic only at the unique slicing scale $2/\sqrt{3}$ the gnomonic image of the hole fixes, so the three roots are the three preimages of $3w$ [14]. Not fitted—forced.

- *Representation theory, the kind.* Of the residue's full $D_6 = S_3 \times \mathbb{Z}_2$, only the $\mathbb{Z}_2$ is a substrate isometry (the parity R, acting on the cut spinor as γ^5), so chirality descends *gauged*; the S_3 is the monodromy (Weyl) symmetry of the solution space and no isometry ($\mathfrak{su}(3) \not\subset \mathfrak{so}(5,1)$), so a family symmetry is *global*—the Standard Model's own arrangement, gauged chirality and global flavour, not assumed but following from which factor is an isometry [26, 24, 15]. The audit behind this count, its receipts, and the splits that cut its law are collected in the programme's combinatorics ledger[L6].
- *Geometry again, the family as vantage.* The S_3 is realised concretely as three 120°-separated hinged vantages: shifting the sky angle by $2\pi/3$ fixes $2M$ and cycles the designated root through all three (the cyclic $\mathbb{Z}_3$), σ swapping the other two within a hinge—three readings of the one substrate, converging from the geometry with the representation-theoretic verdict that the family factor is a description structure, not field copies [14].

The breaking is the slicing's own (the r_0-singling 2+1 factorisation), the mass *values* the ordinary route. **The step that was open when this reading was first stated—the descent—is now built:** whether a spinor sector inherits this three-fold count, that is, whether three *vantages* of the one substrate are the three physical *generations* a single observer sees. That construction is built [28]: a spinor on the slicing structure binds exactly one chiral zero-mode at each throat wall *as a bound mode of the existent leaf* (normalizable in the leaf's proper measure, whereas in the conserved spacetime Dirac norm the same static mode is not)—its chirality the parity $R = \gamma^5$—and the maximally-symmetric matter construction, the one this programme's own principle selects over any single-hinge truncation (which would carry an unfixed arbitrary modulus), places a wall at each of the three hinges. The count is three, and the S_3 permuting the walls is the family symmetry. For the *discrete flavour structure*—the generation count, the chirality, and the family symmetry—the physical identification is therefore a result, forced *within* CR: it rests on the maximal-symmetry principle that defines the programme and on the fermion sector being set by the matter construction, which the wall calculation makes concrete, and on nothing further. Beyond the discrete skeleton the propagating spinor field sector is now built as well [23], the leaf-bound modes and the propagating field being two sectors rather than one owed; what stays the ordinary route is the gauge content, excluded from the continuous isometry [26], and the mass spectrum (electroweak). Bold in shape, and, for the discrete skeleton, a result rather than a conjecture.

One principle, worn three ways. Step back from the descent to the shape of the whole. The three-fold built there [28] is the innermost of three nested registers of maximal symmetry the construction carries. The substrate is maximally symmetric—dS_5, its isometry $SO(5,1)$ exhausted (Prop. 1). The evolving three-space it presents is maximally symmetric—the closed S^3 of the cosh-law cosmology, $SO(4)$ [19, 25]. And the discrete symmetry the slicing breaks out of the substrate is itself maximally symmetric—the hexagonal $\mathrm{Aut}(A_2) = D_6$, the $\mathbf{3} \oplus \bar{\mathbf{3}}$ worn on the two horns, the fundamental $\mathbf{3}$ matter and its R-image $\bar{\mathbf{3}}$ antimatter, at one weight within CR [28, 26]. Read this way the candidate synthesis widens past the generations: the Standard Model's apparent arbitrariness—why three families, why a chirality, why any discrete family structure at all—is, with the descent built [28] and forced within CR, not a list of inputs but the shadow of one principle broken symmetrically: a maximally-symmetric breaking of a maximally-symmetric substrate, carried in a maximally-symmetric space. The grounded floor is the three registers, and the discrete structure they carry is counted and built [28], not conjectured: three chiral families related by S_3, forced within CR. What is *not* claimed—and stays the ordinary route—is a geometric origin for the gauge content or the masses; those are excluded from the isometry, and electroweak. For its discrete flavour skeleton, then, this is the thing the one principle reaches—within CR, at the weight the matter paper states in full.

The two readings: general relativity and the Standard Model's flavour, one discrete structure. Stated at its sharpest, the synthesis is one claim about one discrete structure read two ways. The residue $\mathrm{Aut}(A_2) = S_3 \times \mathbb{Z}_2 \cong D_6$, read as *cuts* of the substrate, is the organising symmetry of general relativity's symmetry-reducible solution space [20]—the three horizon roots, the vantages that permute them, the mass-reflection R, the two null rulings; read on a *fermion sector* it is the Standard Model's discrete flavour skeleton—a threeness against two chiralities, an S_3 and the parity $\gamma^5 = R$ [28]. *The matter sector distinguishes two threes here and this paper follows it*: the S_3 among these vantages is the *within-state* index, and the generations' threeness is the turnaround's deck $\mathbb{Z}_3$ (*ibid.*). One structure, two readings, one slicing operator: the vacuum sector and the flavour skeleton are not two theories to be reconciled but two perspectival shadows of the one substrate's discrete symmetry—the reading, not the read, in the exact sense of §7. Two boundaries hold this at its weight, and both are load-bearing rather than caveats. *What the two readings share is the discrete residue, not the continuous symmetry:* the continuous isometry $SO(5,1)$ generates the cuts and is excluded from the Standard Model ($\mathfrak{su}(3) \not\subset \mathfrak{so}(5,1)$ [26]), so it is the residue the exhausted continuous isometry leaves that both readings share, the gravitational reading keeping the continuous part to itself. *And it is the flavour skeleton, not the gauge structure:* the Standard Model's defining $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ is exactly the part *not* read off the substrate [26], the mass spectrum stays electroweak, and the reading delivers the generation count, the chirality, and the family symmetry and those alone. Nor does the synthesis unify the quantum with gravity: what the two readings share is a discrete *geometric* symmetry, neither quantum nor classical in itself, while "the quantum" enters the corpus separately, in the horizon-thermal register of the ledger ($\hbar$ the Gibbons–Hawking gauge, §6.1)—so the old divide this synthesis bears on is between general relativity and the Standard Model's discrete content, not between general relativity and the quantum. A gravitational–cosmological unification whose matter reach is the discrete flavour skeleton, asserted nowhere as more. The parts are established across the corpus [14, 15, 19, 23], and the two-reading statement is their consolidation.

8 The space the corpus lands in

This section collects, paper by paper, where the geometric core reaches into each of the sixteen and where each reaches back—the connections that belong to no single paper, gathered in one place:

- **P1, the metric singularity [12]** — forward: P1's scope note now points here, reading its metric-singularity boundary as the structural forcing on which the substrate's everywhere-real boundary rests (P1 standing on general relativity alone, needing none of the substrate ontology). Reciprocal, into §4 and the augmentation rung of §6: P1 supplies the *structural* forcing complementary to P4's empirical one — the event horizon is a *metric singularity* (a null hypersurface along whose generators the spatial extent also contracts to zero), occurring only as the null future boundary approached at infinite exterior time, so the layered, cosmically-timed ontology the augmentation reads is forced structurally as well as measured. It is the finite-curvature species of the boundary genus (the infinite-curvature species and the genus in P2), the real boundary of §4 at its first exemplar. The two forcings — structural (P1, from general relativity alone) and empirical (P4, measured) — are independent and co-equal, neither resting on the other.
- **P2, the Schwarzschild circle [13]** — forward: P2's introduction now points here, its two-species genus and perspectival curvature the foundational-geometry instance of the

substrate's everywhere-real boundary and shadow-reading. Reciprocal, into §4 and §7: P2 supplies the *genus* — the event horizon and $r = 0$ are the two r-poles of one homogeneous circle, metric singularities of one genus differing only at second order (curvature), so P1's finite-curvature species and the conjugate infinite-curvature species are one object; and the curvature divergence at $r = 0$ is the curvature of the *perspectival* metric, real as that metric's and reading the chart's labelling — the first, sharpest instance of the shadow-reading of §7 (a real feature of a vantage-dependent description, built-and-real, not an artefact in the pejorative sense). The everywhere-real continuation through the seam (§4) is the circle's own analytic completion. Sbierski's inextendibility concerns the maximal analytic Schwarzschild spacetime, the $\Lambda = 0$ object this construction does not contain; on the substrate the geometry runs through the locus the chart labels $r = 0$, C^∞ across it [18].

- **P3, the slicing curve [14]** — forward: P3's substrate-and-cut section now carries the substrate-level statements this paper owns — the Lorentzian signature intrinsic to the positive curvature, the straight null rulings its own "ruling legs" run along, and the equilateral-pseudo-sphere c–Λ mechanism — pointing here for the full development. Reciprocal, into this paper: P3 supplies the *worked instances* of the core — its slicing curve is the cut of §7 realised; its `prop:flip` and conjugate branch are the seam instance of §4 (the flip confined to the 2-surface, the conjugate region a real branch of the one Lorentzian substrate); its two-sweeps/vantage-swap and its α-invariant/M-perspectival result are the shadow-reading (§7) and the constant-locking (§3) at their first concrete rung. What P3 still hands forward: the coupled operations it now collects and owns—the root-exchange σ and the backward-radial reflection R, coupled across both regimes, generating the substrate's full discrete symmetry—are the geometric base this paper reads at substrate altitude (the orientation $\mathbb{Z}_2$ of §7 being the R-factor of that coupled whole, advanced group-theoretically by P5); and the overcritical/lap conjugacy is the geometric base of the phase-structure-at-the-seam (the geometric base is established; the reading itself is a target rather than a result) (§9).
- **P4, the modern parallax [16]** — *round complete, both ways.* Forward: P4's exclusion section now points here, reading the augmentation it forces as the empirical rung of the substrate's maximal symmetry (carrying its own Copernican scope). Reciprocal, into §6 (the first and fifth rungs): P4 supplies the *empirical* rung — the redshift-isotropy floor (accumulated-redshift scatter $\sim 10^{-3}$ against an observed $\lesssim 3 \times 10^{-6}$, a three-order exclusion) forces the cosmic foliation as a *measured* feature, and `thm:augmentation` establishes CR as the *necessary and sufficient* augmentation of general relativity for a description of an existing, evolving world (the lapse the objective rate of advance, the shift the relativity of synchrony). So the necessity of the first rung is not posited but *measured* — the empirical counterpart of P1's structural forcing, which stands on causal structure alone and needs none of this evidence. Under the Copernican principle the same datum reaches the maximal symmetry of the slices (the fifth rung, universality as the invariance of the measure), which P4 marks as an *extrapolation* beyond the directly constrained region — a scope §6 inherits, not softens.
- **P5, the description groupoid [15]** — forward: P5's `rem:orientation` now reads its orientation-parity $\mathbb{Z}_2$ through this paper's lens — the discrete residue the exhausted continuous isometry leaves unspent, and, not claimed, the structurally indicated matter home — pointing here. Reciprocal, into §7: P5 supplies the residue's concrete realisation — the $\mathbb{Z}_2$ is the SdS solution space's own $\mathrm{Aut}(A_2) = S_3 \times \mathbb{Z}_2 \cong D_6$—the coupled operations the slicing paper owns, advanced here into their group-theoretic form—the mass-reflection $2M \mapsto -2M$ realised as the reticle reflection on the horizon-cubic roots (`rem:orientation`), and the su(3)-guard from the gravitational side (`rem:a2-distinct`:

this A_2 is su(3)'s abstract root system in a *different realisation*, no colour via a continuous isometry — whether the shared abstract type is significant left open). The discrete residue of §7 is thus not an abstraction but a worked group acting on a concrete family.

- **P6, the shadow of existence [17]** — forward: P6 now points here (from its closing placement in the programme), its theory-choice discipline the epistemic ground on which this paper's unification is held as a conjecture rather than an established result. Reciprocal, and it fixes the *altitude* of the whole paper: P6 supplies the *epistemic engine* — the shadow-reading (infer the world that casts the appearances) of §7, the four rules with *require over permit* (Rule 2) the load-bearing one, the modal fallacy, and the constructive ordering (ontology from evidence). The unification of §6 is a rule-favoured structure, and reading its several established rungs as one fact is a theory-choice, decidable by the test of §9 as the discipline's first programme.

- **P7, the framework [18]** — *round complete, both ways.* Forward: P7's null-boundary-correspondence section cites this paper for the intrinsic-Lorentzian result and the c–Λ mechanism, and its synthesis section now points here for the maximal-symmetry reading that gathers its structural recoveries. Reciprocal, and it is the deepest of the rounds: P7 supplies the *established* rungs of §6 and the discipline that holds them. Its central theorem (`thm:augmentation-p7`, `thm:cosmogenesis`) is the necessity-and-sufficiency rung as a formal result; its synthesis reads three standing features of general relativity as one substrate's — the Carter constant [31] *as the substrate's own maximal symmetry surfacing*—a sentence that is true for a reason worth stating, since on Kerr the same words are false: there the Carter constant comes from an *irreducible* rank-two Killing tensor, one provably not built from the Killing vectors, which is why Carter's separation was a discovery rather than bookkeeping. *On a maximally symmetric space the theorem runs the other way*: constant curvature leaves no room for an irreducible Killing tensor, so every one is a symmetrised product of Killing vectors and a quadratic first integral cannot be independent information[9L13], the problem of time dissolved, the Dirac constraint algebra as the symmetric-space grading $[\mathfrak{m}, \mathfrak{m}] \subset \mathfrak{h}$ of $SO(5,1)/SO(4,1)$ — which is the maximal-symmetry unification already at work at the structural rung. P7's synthesis section carries the same reading at the structural rung, where general relativity's solution space closes on one substrate; the unification here is likewise a conjecture, held to the two tests the programme keeps open.

- **P8, the slicing operator [19]** — forward: P8's introduction now points here, its covariance-of-geometries substrate the geometric core's substrate. Reciprocal, into §2 and §7: P8 supplies the *operator* that realizes the substrate's cuts — the slicing curve promoted from classifier to a generating operator for the whole static spherically-symmetric sector, with two derivations at the heart of the core's picture: *straight cuts are vacuum* (the SdS family is exactly the kernel of the matter functional) and *matter is the bend of the cut* ($\rho = m'/4\pi r^2$, the curvature of the cut off the vacuum profile). The covariance is lifted one level — the de Sitter substrate held invariant under change of *geometry* as general relativity holds one geometry invariant under change of chart, "many geometries, one substrate" — the operational form of §2; the reading grounded on the results, not entailed by them (P6).

- **P9, the range [20]** — forward: P9's introduction now points here, its range the reach of the substrate's maximal symmetry. Reciprocal, into §6 and §7: P9 supplies the *reach* of the construction — the range of the slicing operator is exactly the symmetry-reducible sector of general relativity (every reachable geometry carries a sweep-subgroup of $SO(5,1)$), filled across all Petrov types [32], so what the substrate's maximal symmetry reaches is the constrained, non-radiative skeleton of general relativity; the *wall* is where continuous symmetry is lost — genuinely inhomogeneous matter, the onset of free gravitational radiation — the matter-boundary face (P13) read from the range side. And the Carter constant general relativity carries as an unexplained gift is *the substrate's maximal symmetry surfacing* in the separable corner, one of the maximal-symmetry recoveries the unification gathers (with P7). The reachable catalogue is one substrate read through a vantage groupoid over a moduli family of cuts.

- **P10, canonical time** [**22**] — *round complete, both ways.* Forward: P10 now points here (from its dissolution section), its dissolution of the problem of time the canonical face of the substrate's maximal symmetry. Reciprocal, into §6 (the algebroid rung): P10 is the *canonical* complement to P12's structural account of the problem of time — where P12 identifies the obstruction (the "wrong-sign" structure function *is* the substrate's coset metric, its base-dependence the obstruction), P10 supplies the *resolution*: selecting the cosmic foliation the augmentation forces (empirically by the CMB [P4], structurally by the horizon [P1]) deparametrizes the frozen constraint into a *true Hamiltonian*, the frozen and evolving readings the same canonical content differing only in whether the manifold is granted existence — so the problem of time is *dissolved*, not solved, by the same ontological correction the whole augmentation makes. The closed-S^3 graviton tower deparametrizes to unitary cosmic-time evolution and projects to the flat-ΛCDM graviton, matter the bend of the slicing, not ontological content.

- **P11, why the cut bends** [**23**] — *round complete, both ways.* Forward: P11's chirality section now points here, the graviton's handedness the radiating-sector face of the substrate's discrete residue. Reciprocal, into §7 and the wall of §6: P11 supplies the *dynamics* — why and how the cut bends in time (the confined Gowdy–de Sitter wave, its energy and momentum the shear of the leaf, deparametrized to P10's true Hamiltonian), located at the Type-I edge, the last confined stratum before the *wall* — the generative boundary (a Type-N plane wave, *not* a metric singularity) where generation-by-symmetry hands off to ordinary inhomogeneous evolution, the dynamical face of P9's range boundary. And the chirality: past the wall the graviton's two helicities settle a genuine handedness (the turning of the polarization plane), the substrate's *orientation parity* surfacing in the radiating sector — the disconnected $\mathbb{Z}_2$ of §7, beyond the index obstruction (P13), the dynamical face of matter's one geometric residue. Cosmic no-hair makes the de Sitter background an attractor; the continuous dynamics *admits* rather than forces a quantum structure, any forcing isolated to the discrete root structure (P12).

- **P12, the constraint algebroid** [**24**] — forward: P12's scope section now points here, reading its symmetric-space recognition as one rung of the substrate's maximal symmetry (at the coherence weight; its own result standing on the symmetry-reducible sector). Reciprocal, into §6 (the sixth rung) and here: P12 supplies the *deepest structural* rung — general relativity's own hypersurface-deformation (Dirac) algebra *is* the symmetric-space grading $[\mathfrak{m},\mathfrak{m}] \subset \mathfrak{h}$ of $SO(5,1)/SO(4,1)$ (`prop:closure`, term for term), and the "wrong-sign" structure function at the root of the problem of time *is* the substrate's Lorentzian coset metric, its base-dependence the obstruction — so the problem of time dissolves into a single geometric fact of the maximal symmetry, established on the symmetry-reducible sector. The orientation-parity $\mathbb{Z}_2$ of §7 is realised here as the central inversion / double-ruling swap, and the su(3) exclusion carried with its own scope ($SU(3) \not\subset SO(5)$; the A_2 the discrete Weyl shadow only, no continuous colour isometry).

- **P13, the matter boundary** [**26**] — *round complete, both ways.* Forward: P13's closing programme-significance section now points here, reading its wall as one face of the substrate's maximal symmetry. Reciprocal, into §6 (the fourth rung) and §7: P13 supplies the *matter-boundary* rung — the continuous matter symmetry is excluded because $SO(5,1)$ is *complete* ($\mathfrak{su}(3) \not\subset \mathfrak{so}(5,1)$; the geometric-isometry route closed four ways — the σ-lift a real Weyl reflection not the Wick rotation, the cascade dropping the gauge at the real cut, the A_2 skeleton discrete-only, and the Atiyah–Hirzebruch index rendering a compact-face fermion sector vector-like), so the exhausted symmetry that locks the constants leaves no room for a geometric colour. The mechanism is *complementary*: the index acts on the connected gauge group, the gravitational chirality on the discrete orientation parity $\mathrm{O}(5,1) \setminus \mathrm{SO}_0$ the index cannot reach, so observed fermion chirality is *forced* non-geometric — and that same discrete parity is the discrete residue of §7, the single geometric opening the wall leaves The structure is established; the reading of it offered here is not claimed. The compact (Wick) face is real-by-construction but, atemporal, no co-equal existent, by CR's existence criterion.

- **P14, the fermion sector** [28] — forward: P14 points here, reading its three chiral generations as the innermost of the three registers of maximal symmetry — the maximally-symmetric *discrete* breaking the slicing takes from the substrate. Reciprocal, into §7 and the matter rung of §6: P14 supplies the *built* discrete residue — the boundary's single geometric opening realised, three chiral families related by a global S_3 with $R = \gamma^5$ chirality, forced within CR as the maximally-symmetric slicing of a maximally-symmetric substrate, the gauge and mass content external. The count is the substrate's discrete symmetry worn as matter, originating at the S_3-fixed Nariai crest and unfolding as the universe expands; whether it coheres with the empirical Standard Model stays the data's to judge.

- **P15, the cosmology** [25] — *round complete, both ways.* Forward: P15's introduction points here for the everywhere-real substrate-and-boundary ontology, and its discussion for the root of its rigidity — Λ the maximally symmetric substrate's sole scale. Reciprocal, into §6 (the third rung): P15 supplies the *observational* rung — with Λ the sole scale the cosmology carries no tunable freedom (the parameter-free low-multipole deficit set by Λ through r_0 and D_C; the geometric rate; the transmission dichotomy — no inflationary scale-invariant attractor, no consistency relation, no substrate-sourced B-modes), its discriminating power now returned on the data — the genuine-Boltzmann low-multipole shape a cosmic-variance-limited wash (a smooth deficit bottoming at $\ell = 4$, neither the sharp observed quadrupole dip nor a sharp falsification), while the geometric rate resolves the Hubble tension and the primordial abundances return deuterium and helium within measurement on the corpus's own hot dense history — the one measured initial condition ρ_r/ρ_m an η-analogue read not tuned. The altitude discipline still holds §6 at coherence, but the observational rung P15–P16 supply stands above it: on the discriminating axes the cosmology has begun to earn correspondence, so that "coherence shown is not correspondence earned" now marks the boundary between the structural unification (held, still, at coherence) and the empirical confrontation (where correspondence has started to be paid), not the whole programme's ceiling. The cosmic beginning is the cosmogenesis branch point—the substrate finite in curvature across it, the geometry read over the areal chart divergent there—reached through the conjugacy yet real.

- **P16, the cosmogenesis** [27] — *round complete, both ways.* Forward: P16 points here, reading the Big Bang as a synthesis of the corpus's deductively forced consequences over this maximally symmetric substrate — the hot dense era the far side of a collapse the substrate's own geometry already carries through the branch point, finite in the substrate's curvature whatever the areal chart reports there. Reciprocal, into §4 and §7: P16 lands the *lap* — the conjugate branch of the slicing curve, reached through the imaginary yet real, read forward as a previous universe's collapsed matter, the *cosmogenetic bead* the framework proves one closed curve [18] and a concrete instance of this paper's reached-through-the-imaginary principle (its cosmic time runs off the real axis through a bounded contour while the areal radius stays real at every point it lands on) — and returns the one quantitative consequence the synthesis adds beyond the corpus's established results: the infalling matter, heated above the deuterium bottleneck and re-expanded through it at the standard rate, runs a standard nucleosynthesis that *produces* the primordial light-element abundances (helium-4 and deuterium at their observed values, the lithium problem shared with ΛCDM). The Big Bang is thus no initial edge of the substrate but a broken-symmetry shadow cast within it — the substrate's maximal symmetry read at the cosmogenesis rung, the abundances its fossil; its open edge whether one progenitor handover fits all abundances together.

9 Frontiers

1. **The free-data-count test of the unification.** Show that the framework's count of genuinely free data equals the substrate's non-symmetric residue—maximal symmetry, spent, leaving exactly the one measured cosmological IC (ρ_r/ρ_m) as the sole tunable datum, and each constant-relation a locked parameter. Decisive: a hidden geometric freedom in the cosmology, or a genuinely free constant where the reading says it should lock, falsifies "rigidity and wall are one fact." The item has two sides and they now stand differently, so it is split rather than carried whole. *The CONSTANT side is a result and this paper carries it*: §6.1 states that the gravitational–cosmological–quantum sector spends *no free dimensionless constant*, and the reason is structural rather than counted—neither real form supplies a second invariant and a dimensionless magnitude needs two[P17R13], so there is a single dimensionful input and nothing for a dimensionless tuning to hide in. *That argument is about the geometry's own invariants, and it does not reach the regularisation of a mode sum*: the graviton tower's log-scale coefficient is computed and is not zero [22], so one dimensionless constant is spent there. *The structural reason above survives intact — there is still no second invariant — and what it does not establish is that no constant enters by any route at all. And the falsifier's second clause was met rather than asserted*: a place where a genuinely free constant could have lived was checked and it locked—the metric function, for a time the one input the construction did not derive, follows entirely from $T^t{}_t = 0$, a first-order linear equation whose whole solution space is the Tangherlini–de Sitter family with M the single constant of integration [14], obtained rather than assumed[P3R34]. *The DATUM side remains a target*, and it is a different kind of quantity: ρ_r/ρ_m is a matter initial condition and not a constant, so it is not in the geometric residue at all, and the programme calls it a one-parameter accommodation. *And what was written here as owed is a result, which is a different thing*[9]: the derivation from the progenitor collapse cannot be delivered, for a structural reason rather than for want of work. Radiation dilutes as a^{-4} and matter as a^{-3}, so $\rho_r/\rho_m \propto 1/a$—*a quantity that changes along the leg has no single value for a handover to transmit*, and the progenitor's value at its own maximum is a reading taken at one point on a curve. Nor does the crossing supply one: a factor multiplying both components alike leaves the ratio unchanged, and only a species-resolved factor could rescale it—which is exactly the rule withdrawn when the crossing was shown lossless for every species [18]. *So ρ_r/ρ_m is where the observable leg's own clock is read*, and a clock's zero is not the sort of thing a handover carries. *A frontier closed by impossibility is a finding and this list records it as one*: what stands here is not a derivation owed but the statement that there is none to give. *And the companion cosmogenesis paper reaches the same line from the other side, on a different ground*: what a handover carries is what a conservation law protects, so η crosses the compression peak because baryon number is conserved through

it, while the progenitor's composition is erased there because nuclear binding is not—"two kinds of thing," in that paper's words [27]. *Two independent reasons, agreeing*: the ratio is not carried because it is not constant along either leg, and it is not carried because the crossing destroys what would have fixed it.[P17R16]

2. **The maximal-symmetry ledger.** Systematically determine what geometric relations among the fundamental constants the substrate forces (c–Λ and Nariai the two known instances), reading each as a spent parameter; and look for any constant maximal symmetry does not reach—the constant-side analogue of the discrete-parity matter opening. This is grounded in two proven instances.

A third item stood here and has been decided, in both halves, so it is recorded as a result rather than carried as a frontier. The phase structure at the seam is *real structure and not interpretation* at the level of the object: the antilinear face K and the reflection R are the two axis-symmetries of one analytic object, whose composite supplies every kinematic datum of charge conjugation, with the self-conjugate photon congruence as K's fixed set; the reading is neither a second labelling of matter and antimatter, which R already performs linearly, nor a continuous interpretive parameter, reality admitting exactly two values. *And it does not attach to trajectories*: a mass enters the mode equation as m^2a^2, and the branch point annihilates that term while $|aH|$ diverges, so $\omega/|aH| \to 0$ independent of m—a massive trajectory carries no phase, by failing to freeze rather than by freezing into one[9].

Appendix R Computational Receipts

Each computation marked R in the text is verified by a runnable receipt in the `receipts/` directory that ships with this work; status OK = verified (traced, run, output matches the claim). This appendix is generated from the receipt index, not maintained by hand.

P17R2 `P17_the_alpha_family_is_a_foliation` [OK]
the substrate ontology / the c–Lambda null-ruling gauge (THE ALPHA-FAMILY IS A FOLIATION OF THE AMBIENT BY THE LEVEL SETS OF ONE QUADRATIC FORM, WITH THE LIGHT CONE AS ITS SINGULAR MEMBER – AND THE CONFOCAL ORTHOGONALITY THEOREM, WHICH A STANDING LEAD EXPECTED TO APPLY, IS VACUOUS HERE FOR A STRUCTURAL REASON. THE PREMISE IS CORRECT: in the EQUILATERAL case – which is what this substrate is – the confocal family IS the dilation family. ** BUT THE CLASSICAL THEOREM IS NOT 'CONFOCAL QUADRICS ARE ORTHOGONAL' FULL STOP: it is that through a generic point pass THREE members, one of each type, and THOSE meet pairwise orthogonally. The confocal equation is CUBIC in the parameter generically and LINEAR in the equilateral case – asserted – so exactly ONE member passes through any point and there is no second member for it to be orthogonal TO. The hypothesis fails, not the conclusion. ** The same fact without machinery: a point assigns ONE value to the form, so distinct-alpha substrates are DISJOINT and cannot meet at any angle. ** THE DEGENERATION, NAMED: equilateral is exactly the confocal family's degenerate case – as the semi-axes come together the cubic loses two roots, the three types collapse, and the triply-orthogonal system collapses with them. So the theorem went unused for the BEST POSSIBLE REASON: this substrate sits at the one point of the confocal moduli space where it has no content. ** ** AND WHAT SURVIVES IS BETTER THAN THE THEOREM WOULD HAVE BEEN: the family is a foliation – de Sitter above zero, THE LIGHT CONE at zero, the two-sheet hyperboloid below. Different-alpha substrates NEST, do not meet, and share ONE asymptote, which is the family's own degenerate member. **). *Computes:* the equilateral reduction computed; the degree in the confocal parameter computed and asserted (3 generic, 1 equilateral); the three signature classes tabulated *Bound:* ** AND THAT IS A REASON FOR THE c–Lambda NULL-RULING GAUGE'S alpha-INDEPENDENCE, FROM THE FAMILY RATHER THAN FROM THE GAUGE: the null structure is carried by the one leaf common to the whole family, so it cannot depend on which leaf one stands on. This paper already calls the cone 'the locked asymptote' without connecting it to the family it belongs to. AND ONE ROUTING THE LEAD DID NOT ASK FOR: the map between members is a DILATION, which is CONFORMAL and not an isometry, so the alpha-family is a CONFORMAL ORBIT and belongs in the conformal ledger rather than the quadric-geometry file where it was left – an item filed under the machinery someone first reached for rather than the machinery that fits it ** *Run:* `python3 receipts/P17_geometric_core_paper/P17_the_alpha_family_is_a_foliation.py`

P17R3 `P17_power_of_a_point` [OK]
sec:power (**!! c54.161: THE CIRCULARITY IS RETIRED** – the power is now MEASURED from bisected secant products and compared against X0 taken from the substrate constraint, at 2 alpha and 3 alpha, so eq:powerisheight is no longer true by construction. **!! MARKED c54.157: THE CITING SENTENCE ATTRIBUTES TO THIS FILE FIGURES IT DOES NOT PRODUCE** – the ds^2, the 600 random heights and the max 5.7e-14 all come from 'P17_power_is_null'. **And its own eq:powerisheight is circular: X0 is DEFINED as sqrt(|X|^2 - alpha^2), so 'power = X0^2' holds by construction.** What it does compute honestly is Euclid III.36 – the secant invariance to 4.4e-16 over 500 sightlines). *Computes:* every-sightline invariant test at machine precision *Run:* `python3 receipts/P17_geometric_core_paper/P17_power_of_a_point.py`

P17R4 `P17_power_is_null` [OK]
sec:power prop:tangentnull (tangent to throat is NULL: tangent length=|X0|=sqrt(R^2-alpha^2) identity (Euclid=hyperboloid RHS); ds^2=-dX0^2+|dX|^2=0, max 5.68e-14 over 600 heights (=paper's 5.7e-14); power-of-a-point = null condition, same equation). *Computes:* tangent-length identity + ds^2=0 at 600 random points *Run:* `python3 receipts/P17_geometric_core_paper/P17_power_is_null.py`

P17R5 `P17_qm_S4_vs_S5` [OK]
sec:ledger (**!! CORRECTED c54.157: THE ROW CLAIMED THREE VERIFIED ITEMS AND ONE WAS VERIFIED.** The realified-generator antisymmetry is computed; **'minimal faithful real rep R^6' was a print string** (exhibiting one 6-dimensional rep shows su(3) FITS in so(6) and

excludes nothing about 5), and 'GH beta = 2 pi alpha n-independent' has kappa hardcoded to 1.0 and never varies with n. **The minimality is now ENUMERATED by the Weyl dimension formula over su(3)'s irreps: the smallest faithful real dimension is 6, attained by 3 + 3-bar, so no faithful real rep fits in R^5.**). *Computes:* realified-generator antisymmetry + faithful-dimension + GH n-independence *Run:* `python3 receipts/P17_geometric_core_paper/P17_qm_S4_vs_S5.py`

P17R6 `C3_inversion_extends` [OK]
rem:onecircle (C3 — does INVERSION in the throat extend off the equatorial plane, as polarity did (Q1/Q6)? Minkowski inversion in the quadric eta(X,X)=alpha^2: P* = alpha^2 P / eta(P,P).). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P17_geometric_core_paper/C3_inversion_extends.py`

P17R7 `D1_hinge_duality` [OK]
rem:onecircle (D1 — PROJECTIVE DUALITY: the dual of the swinging door. PRIMAL : the pencil of slicing planes (doors) through the HINGE. DUALITY : in the plane of the throat, w.r.t. the throat circle |x|^2 = alpha^2, the pencil of LINES through a point P dualises to the RANGE OF POINTS on P's POLAR. So the dual of the swing is a POINT SLIDING ALONG THE HINGE'S POLAR. Where is that line?). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P17_geometric_core_paper/D1_hinge_duality.py`

P17R8 `D3_segre_no_pencil` [OK]
rem:onecircle (D3 — SEGRE. The Segre symbol classifies a PENCIL of quadrics by the elementary divisors of lambda A + mu B. So the probe needs an honest answer to: DOES THE CORPUS HAVE TWO QUADRICS IN ONE SPACE? Confirmation 4, before anything runs. Candidates, each checked:). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P17_geometric_core_paper/D3_segre_no_pencil.py`

P17R9 `O2_sightline_null_on_lift` [OK]
prop:tangentnull (O2 — p0's tangent/secant distinction, and whether it is a shadow boundary in the optics sense. p0: 'sightlines of the projection that TOUCH the throat are light rays; those that CUT it are not.'). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P17_geometric_core_paper/O2_sightline_null_on_lift.py`

P17R10 `Q1_quadric_polarity` [OK]
rem:onecircle (Q1, invariant form. The first pass computed a EUCLIDEAN distance for the polar hyperplane in a MINKOWSKI space and got 1/sqrt7 = 0.37796 – a gauge artifact, not an invariant. Redone with the only quantity polarity actually depends on: eta(P,P) against alpha^2.). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P17_geometric_core_paper/Q1_quadric_polarity.py`

P17R11 `Q3_cayley_klein` [OK]
— (Q3 — Cayley-Klein: is the substrate's metric a log-cross-ratio against the absolute, and is alpha the CK scale constant? Substrate dS_5 = {X in M^6 : eta(X,X)=alpha^2}.). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P17_geometric_core_paper/Q3_cayle`

P17R12 `Q6r_polar_is_the_background` [OK]
— (Q6-CONFIRMED — the polar slice sits at RUNG 2, and that answers A18. CONFIRMATION 1 (ontology): section 0's five-rung ladder. Rung 1 = the substrate dS_5 = SO(5,1)/SO(4,1). Rung 2 = 'A CUT of the substrate – a four-geometry. dS_4 is one such: the maximally symmetric one... A geometry is a cut exactly when its isometry group contains a sweep-subgroup of the substrate's.'). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P17_geometric_core_paper/Q6r_polar_is_the_background.py`

P17R13 `P17_no_second_scale_on_either_face` [OK]
the constant ledger (**NEITHER REAL FORM OF THE SUBSTRATE SUPPLIES A SECOND

SCALE, SO THE CONSTRUCTION CANNOT FORCE A DIMENSIONLESS MAGNITUDE. THE SILENCE IS A PROPERTY, NOT A GAP.** The Wick rotation X0 -¿ i X0 carries the defining quadric to S^5 of the SAME radius alpha – it acts on the COORDINATE and leaves the right-hand side untouched, so no step exists at which a second radius could enter; the two faces are the two real forms of one complex quadric and a real form does not choose its own radius. Every curvature invariant on both faces is a pure power of 1/alpha^2 (asserted by logarithmic derivative), leaving no second constant with which to form a ratio. The remaining candidates are enumerated and each disposed: the horizon period is built from alpha; M and r_0 are moduli; 2/(3 sqrt 3) is one member's value. **So a dimensionless magnitude cannot be forced, and that is the COMMON ROOT of three verdicts already banked separately – the winding quantises and cannot measure, the flat bundle selects and cannot couple, the branch point filters and cannot supply.**). *Computes:* the Wick image asserted equal to S^5 of radius alpha; each invariant's logarithmic derivative asserted an integer *Bound:* **bound: this is about the two faces' GEOMETRY. Fields living on either face may carry their own scales – but those are CONTENT, adjoined rather than derived, and adjoining one is what the corpus's admissibility criterion rejects. The escape exists and costs the thing the programme is for** *Run:* `python3 receipts/P17_geometric_core_paper/P17_no_second_scale_on_either_face.py`

P17R1 `p0_the_entropy_declination_is_load_bearing_for_the_ledger` [OK]
'sec:ledger''s entropy declination; 'PO-43', 'PO-32' (**the declination is load-bearing and was not known to be**: a topological term contributes no field equation, so its coefficient's only observable is a Wald-entropy contribution going as $\{}chi(\{}Sigma)$ — **an area-independent constant shift** for a 2-sphere. And the ledger has **no absorber** for a constant entropy shift as it has $\{}Lambda$ for a constant vacuum energy. ⇒ **the two branches disagree**: if $S=A/4$ carries the ledger gains a free dimensionless constant it cannot absorb; if it fails, the coefficient has no home and 'p0''s "a result and not a gap" obtains [borrowed from p0]). *Computes:* the two branches shown to disagree, so the topological-terms question cannot be settled independently; and the absence of an absorber, against $\{}Lambda$'s absorption of a constant vacuum energy which is 'p0''s own mechanism *Bound:* **IMPORTED: the Wald-entropy contribution of Gauss–Bonnet going as the Euler characteristic** (Jacobson–Myers), the only import. NOT claimed: that $S=A/4$ fails — that is the open question and this does not settle it; nor is the a-type coefficient computed *Run:* `python3 receipts/P17_geometric_core_paper/p0_the_entropy_declination_is_load_bearing_for_the_ledge`

`N1_the_entropy_and_the_fine_tuning_factor_are_one_number` [OK]
sec:unification (N1 — p0's IDENTITY CHECKED: THE de SITTER ENTROPY AND THE COSMOLOGICAL-CONSTANT FACTOR ARE ONE NUMBER. Information-theory field bake, probe N1, and the field's one quantitative probe. p0 claims in its own voice that the Bekenstein-Hawking entropy 3pi/(Lambda l_P^2) and the fine-tuning ratio 8pi/(Lambda l_P^2) 'differ by 3/8 and by nothing else', so the 10^122 of the one is the 10^122 of the other. Checked symbolically: S = A/4l_P^2 with A = 4 pi alpha^2 and alpha^2 = 3/Lambda gives 3pi/(Lambda l_P^2); rho_Lambda = Lambda/8 pi G with G = l_P^2 against a 1/l_P^4 estimate gives 8pi/(Lambda l_P^2); the ratio is exactly 3/8 and both coefficients are pure numbers, which is the load-bearing half of 'and by nothing else'. At the observed values both are 10^122. Control: the de Sitter TEMPERATURE must not be of that form, or p0's one-register / cross-register distinction would be empty — T carries no l_P at all. The headline had no receipt before this. [borrowed from p0]). *Computes:* runs clean; six verdicts including a control that could have emptied the distinction *Bound:* derived here; a CONFIRMATION rather than a correction — the claim is exact as written *Run:* `python3 receipts/p0_geometric_core/N1_the_entropy_and_the_fine_tuning_factor_are_one_number.py`

P3R34 `P03_operator_at_general_D` [OK]
rem:dimension / thm:kernel (** ASSUMPTION (A1) DISCHARGED. ** CR's OWN operator at general D returns Tangherlini-de Sitter and nothing else. The point previously missed: CR's operator is NOT a metric function – P8 thm:kernel defines the vacuum sector as the KERNEL OF THE MATTER FUNCTIONAL on a cut in the construction gauge ('derived, not matched'), and that condition runs at any D. Computed from the metric rather than quoted: for ds^2 = -f dt^2 + dr^2/f + r^2 dOmega^2_{D-2}, G^t_t = (D-2)/(2r^2)[r f' + (D-3)(f-1)] (verified at D=4,5,6), so T=0 is the first-order linear ODE r f' + (D-3)(f-1) + 2 Lambda r^2/(D-2) = 0 – P8's own equation at D=4 – whose ENTIRE solution space at D=4,5,6,7,8 is f = 1 - 2M/r^(D-3) - 2 Lambda r^2/((D-1)(D-2)) with ONE constant of integration, verified identical to Tangherlini-dS at every D tested. With alpha(D) = sqrt((D-1)(D-2)/2Lambda), which returns sqrt(3/Lambda)

at four, this is f = 1 - 2M/r^(D-3) - r^2/alpha^2 at every D, and the dimension result's input 2M = r_0^(D-3) - r_0^(D-1) is that f at a root. THE DIMENSION RESULT THEREFORE STRENGTHENS: the collapse, the fold being D-1 and the mass parity all run on a function the construction GENERATES rather than borrows [borrowed from P3 / P8]). *Computes:* the Einstein tensor computed from the metric at D=4,5,6 and matched to the closed form; the ODE solved by dsolve at D=4..8 and compared term by term with Tangherlini-dS *Bound:* TWO INPUTS REMAIN AND ARE NAMED: (i) THE LOCK – the construction gauge g_tt g_rr = -1 is assumed at general D, and P8 says the single slicing curve enforces it at four but this does not show it does so at general D (L-89; P8's lapse split is the machinery that would say otherwise); (ii) THE SPHERE – the transverse space is taken round S^{D-2}, which at general D is a choice (L-90). And (A2), the single-harmonic collapse being a REQUIREMENT at other D, is untouched *Run:* `python3 receipts/P03_SdS_slicing/P03_operator_at_general_D.py`

P17R14 `P17_the_cut_is_planar_only_at_M_zero` [OK]
the ladder / sec:slicing / sec:pd (**!! c54.161: THE TWO c54.157 MARKINGS ARE RETIRED** – the hyperplane split is now done for a tilted n with the induced Ricci scalar computed as 12/(alpha^2-c^2), and the Kretschmann scalar is DERIVED from the metric rather than typed in. **!! CORRECTED c54.157: THE ROW'S 'the hyperplane split done in general' IS FOUR PRINT STATEMENTS AND A HARDCODED TABLE**, and the Kretschmann scalar 48M^2/r^6 + 24/alpha^4 is an INPUT typed at line 97, not a finding – it is derived from the metric in 'P07_sds_kretschmann', which is where the paper should look. The remaining evidence items (the M-factored closure defect, the r-dependent kappa^2, the M = 0 control) are accurate). *Computes:* the hyperplane split done in general; the Kretschmann derivative; the two determinations of G' computed and subtracted; kappa^2 solved for and shown r-dependent; the M = 0 control returning residual 0 and G = 0 (the planar vacuum cut) *Bound:* ** IT DOES NOT REFUTE THE CUT RELATION, and reading it that way would be the error the receipt exists to prevent: P9 defines a cut by its isometry group containing a sweep-subgroup, P12 by an orientation plus causal-vantage datum, and the framework calls SdS a DERIVATIVE metric, not an induced one. What is established is that the ISOMETRIC-EMBEDDING reading is available at M=0 and nowhere else, so the corpus owes a sentence saying which relation 'cut' names ** *Run:* `python3 receipts/P17_geometric_core_paper/P17_the_cut_is_planar_only_at_M_zero.py`

`U3_the_residue_is_one_and_it_is_already_counted` [OK]
sec:ledger (the non-symmetric residue is one dimensionful parameter: the dilation [borrowed from p0]). *Computes:* dim O(5,1)=15 vs dim(O(5,1)xR+)=16; asserts p0 and P5 state it and that the frontier item still reads open (13) *Bound:* NOT-A-PAPER-CLAIM — discharges L-201 *Run:* `python3 receipts/L200_free_data_count/U3_the_residue_is_one_and_it_is_already_counted.py`

P17R15 `P0_the_order_parameter_is_the_offset_and_it_is_bounded_by_the_nariai_member` [OK]
sec:residue (**ITEM 48 WORKED: THE TWO SYMMETRY BREAKINGS ARE ONE OBJECT, AND JOINING THEM GIVES THE HIGGS IDENTIFICATION ACTUAL CONTENT.** Lead 'L-521'; VEIN 'L-221'. **PART 1** p0 states the parity as $r_0\{}mapsto-r_0$ (whence $2M\{}mapsto-2M$); P3 derives $2M=r_0-r_0^3$ for a different purpose; **the cubic is odd, so p0's mass-reflection is an IDENTITY of P3's relation** and the two breakings are one object. **PART 2** $2M=0$ at $r_0=0,\{}pm\{}alpha$ with $f=1-r^2/\{}alpha^2$ at all three — **the symmetric sector is the bare substrate**, one geometry read at three offsets. **PART 3** over P3's admissible range $\{}lvert r_0\{}rvert\{}le2\{}alpha/\{}sqrt3$, **$\{}lvert M\{}rvert\{}le\{}alpha/3\{}sqrt3$ — the Nariai mass, derived independently from $f=f'=0$**; a quartic potential is unbounded in its field and this order parameter is not, saturating at the configuration a collapse reaches. **PART 4** eight source checks across p0 and P3, including that p0 now names the Higgs (zero uses in seventeen papers) and keeps 'F1' explicit. [borrowed from p0]). *Computes:* **the oddness of $2M(r_0)$ — this IS the join; if P3's relation were not odd, p0's mass-reflection would be a separate assumption and there would be no single object**; all three roots asserted to give the same de Sitter metric function, both as the symmetric-sector claim and so "three roots" cannot be read as three vacua; the bound asserted numerically over P3's own range AND identified with the Nariai mass **derived independently and never substituted in, because the whole content is that two separately-derived numbers coincide**; and eight source checks *Bound:* **scope: a statement about the MECHANISM, which is what item 48 asked for. NOT claimed: any vacuum expectation value, electroweak scale or mass value; any derivation of the Higgs field, its potential or its gauge group. 'F1' untouched and said so in the paper's own voice. And the three roots

are three offsets on ONE geometry — vantage multiplicity already in the corpus.** *Run:* `python3 receipts/P17_geometric_core_paper/P0_the_order_parameter_is_the_offset_and_it_is_bounded_by_the_nar`

`S1_a_massive_mode_freezes_too_and_the_mass_term_vanishes_at_the_branch_point` [OK]
sec:what-crosses (A2 (PO-seam / PO-7 inversion 3): a massive mode freezes too, for ANY mass — the branch point a-¿0 annihilates the mass term m^2 a^2 while |aH| diverges, so omega/|aH| -¿ c_s k x -¿ 0 independent of m; a massive trajectory carries no phase from non-freezing, closing PO-7 route 3 from the massive side [borrowed from P15]). *Computes:* derives the mass term +m^2 a^2 from the covariant action (sympy), shows omega/|aH| at the crossing is m-independent to the digit up to m=1e5 across ell 28-2475, freeze-out epoch unmoved (5 checks) *Bound:* NOT-A-PAPER-CLAIM — A2; informs L-202/L-171; CRPHI stays assigned; F5 unsoftened *Run:* `python3 receipts/L806_massive_freezing/S1_a_massive_mode_freezes_too_and_the_mass_term_vanishes_at_the_bran`

P17R16 `P17_the_frontier_item_is_a_result_and_the_cosmogenesis_paper_reaches_it_the_other_way` [OK]
sec:seven (**ITEM 18: p0's FRONTIER DATUM IS A RESULT, NOT AN OWING** — the derivation of $\{}rho_r/\{}rho_m$ from the progenitor collapse CANNOT BE DELIVERED (the ratio scales as 1/a, so no single value exists to transmit), and P16 reaches the same line by a conservation law. Lead 'L-530'; VEIN 'L-202' [borrowed from p0]). *Computes:* asserts the old owing sentence ABSENT; eight checks on the replacement incl. the 'X1' cite; four that P16's body says what is now attributed to it; and 'peak spacing' counted 0 in P16's body, 0 in p0, 6 in P15 — plus the header-comment provenance of the withdrawn draft *Bound:* NOT-A-NEW-DERIVATION — the impossibility is 'X1''s (r2433) and is cited, not re-run; 'L-202' stays open *Run:* `python3 receipts/P17_geometric_core_paper/P17_the_frontier_item_is_a_result_and_the_cosmogenesis_paper_reac`

P17R17 `P17_the_ds_entropy_is_the_gauge_count_squared_and_is_the_cc_number` [OK]
sec:ledger (**ITEM 52: THE de SITTER ENTROPY IS THE GAUGE-COUNT SQUARED — AND THE SAME NUMBER AS THE COSMOLOGICAL-CONSTANT FACTOR, NOT MERELY THE SAME MAGNITUDE.** $S=\{}pi(\{}alpha/\{}ell_P)^2=3\{}pi/(\{}Lambda\{}ell_P^2)$ against $8\{}pi/(\{}Lambda\{}ell_P^2)$: they differ by $3/8$ and by nothing else. Lead 'L-532'; VEIN 'L-165' [borrowed from p0]). *Computes:* reduces S symbolically to both printed forms and the CC factor to $8\{}pi/(\{}Lambda\{}ell_P^2)$, pinning the ratio at exactly 3/8; evaluates both at p0's own $\{}Lambda\{}ell_P^2$; nine source checks and three anchor checks *Bound:* DO-NOT-ASSERT — that $S=A/4$ carries to a cosmological horizon on this reading is NOT settled; 'PO-6''s one-constant question untouched *Run:* `python3 receipts/P17_geometric_core_paper/P17_the_ds_entr`

`X1_the_ratio_is_a_clock_reading_not_a_carried_datum` [OK]
sec:seven (**'L-150' ANSWERED IN THE NEGATIVE — the programme's longest-standing target cannot be met.** $\{}rho_r/\{}rho_m$ is not the KIND of quantity a handover can carry: it scales as $1/a$ on either leg so no single value exists to transmit, and a crossing multiplying both components alike leaves the ratio unchanged. Lead 'L-150'; VEIN 'L-202' [borrowed from p0]). *Computes:* the $1/a$ scaling on both legs; the multiplicative crossing's cancellation; and the corpus's own withdrawal of the species-selection rule that would have been needed *Bound:* CONVERTS the datum half from an OPEN TARGET into a CLOSED NEGATIVE with a stated reason — weaker than closure; nothing here bears on $z_{\{}rm onset}$. **INDEXED r2566+c54.207 by 54, having sat unindexed since r2433** — the row was missing, not the receipt *Run:* `python3 receipts/L150_the_datum/X1_the_ratio_is_a_clock_reading_not_a_carried_datum.py`

`I50_the_carter_constant_is_the_substrates_symmetry_and_kerrs_is_not` [OK]
sec:shadows, sec:landing (p0 says twice that the Carter constant is "the substrate's own maximal symmetry surfacing", and this measures the theorem under it: on a maximally symmetric space EVERY quadratic first integral is already a product of Killing vectors, so a Carter-type constant carries no information the symmetry did not — while on Kerr it is IRREDUCIBLE, which is what makes p0's sentence a claim rather than a tautology [borrowed from p0]). *Computes:* counts quadratic first integrals on S^n by solving {Q,H}=0 in a truncated basis INDEPENDENTLY of the products (6=6=6, 20=20=20, 50=50=50, and 105=105=105 at n=5, the substrate's own dimension; flat across truncation degrees 2/3/4); verifies the Kerr tensor really is Carter's by integrating a non-equatorial geodesic (drift at the central difference's own eps^(2/3) floor, a random symmetric control at 4.2e-1); then shows it adds a fifth dimension to span{g, xi.xi, xi.eta, eta.eta} *Bound:* SUPPORTS p0 — and REDIRECTS 59's locator row 13, whose predicted reason ("maximal sym-

metry is the largest algebra of first integrals") names the algebra the Carter constant is famous for NOT being in *Run:* `python3 receipts/p0_geometric_core/I50_the_carter_constant_is_the_substrates_symm`

`T50_the_imaginary_route_to_the_horizon_radii_is_forced_and_two_members_escape_it` [OK]
sec:imaginary (p0's section title is "Reached through the imaginary, real everywhere it lands" and its three instances are all places the imaginary is a CONVENIENCE; the horizon cubic is a fourth where it is forced, since over Q(2M) the cubic is irreducible with three real roots below Nariai, which is exactly the casus irreducibilis hypothesis pair, and no root then lies in a real radical extension [borrowed from p0]). *Computes:* asserts both hypotheses symbolically over Q(M) (irreducible; disc = 4 - 108M^2 not a square, so Gal = S_3 over the REAL base too); measures the Cardano intermediate as non-real with every root real; and carries four controls that must go the other way – over-critical 2M=2/5 (disc ¡ 0, one real root, Cardano REAL), r^3-2 (irreducible, one real root, real radical), r^3-r (three real roots, reducible), and 2M=3/8 (an undistinguished under-critical mass whose cubic is ALSO reducible with real-radical roots) *Bound:* SUPPORTS p0 sec:imaginary – the strongest case for the section's law is the one where no other road exists. NOT CLAIMED: that P05's C(2M) Galois computation is wrong (it is right, and correct for monodromy); NOT CLAIMED that the M=0 and Nariai reducibility selects anything – the reducible masses are dense and the receipt asserts the counterexample beside the coincidence *Run:* `python3 receipts/p0_geometric_core/T50_the_imaginary_route_to_the_horizon_radii_is_forced_and_two_members_e`

Appendix L The Knowledge Ledgers

Each claim marked L in the text rests on a field bake or the figure–theorem bake recorded in the named ledger, which ships with this work and carries the probes, the receipts, and the three registers kept apart — what bit, what bounced, and where the boundary is. A ledger is working apparatus, not a publication, and is cited here rather than in the bibliography. This appendix is generated from the ledger index, not maintained by hand.

L5 `CATEGORY_THEORY_LEDGER.md` [field bake]
category theory against CR — the corpus's largest unlisted field

L6 `COMBINATORICS_LEDGER.md` [field bake]
The combinatorics field-bake ledger — what bit, what did not, and why. Lane 8

L8 `CONFORMAL_GEOMETRY_LEDGER.md` [field bake]
conformal / Möbius geometry against the substrate — the field that refused, and why

L10 `FUNCTIONAL_ANALYSIS_LEDGER.md` [field bake]
The functional-analysis / unitarity field-bake ledger — the field that bounced, and the one routing fact it returned. Third of the four fields ‘L-272‘'s re-survey left outstanding. ‘OWED‘ 622

L12 `HARMONIC_ANALYSIS_LEDGER.md` [field bake]
The harmonic-analysis field-bake ledger — what bit, what bounced, and the boundary. Second of the four fields ‘L-272‘'s re-survey left outstanding. ‘OWED‘ 622

L17 `QUADRIC_GEOMETRY_LEDGER.md` [field bake]
projective geometry of quadrics against the CR substrate — the CK metric identification, and the ladder gap it exposes

L19 `SPECTRAL_THEORY_LEDGER.md` [field bake]
The spectral-theory field bake — what bit, what bounced, and the boundary. Tier B's largest never-thrown field ($\times$189 on the reach measure), verified as NOT covered by the harmonic ledger, which mentions spectral, self-adjoint and deficiency zero times

L1 `FIGURE_THEOREM_LEDGER.md` [theorem bake]
The figure–theorem ledger: which classical theorem each figure carries, and its receipts

L21 `INFORMATION_THEORY_LEDGER.md` [field bake]
The information-theory field-bake ledger — what bit, what bounced, and where the boundary is

L16 `NUMBER_THEORY_LEDGER.md` [field bake]
The number-theory field-bake ledger — what bit, what bounced, and where the boundary is

L13 `INDEX_THEORY_LEDGER.md` [field bake]
The differential-topology and index-theory field-bake ledger — what bit, what bounced, and where the boundary is

L14 `INTEGRABLE_SYSTEMS_LEDGER.md` [field bake]
The integrable-systems field-bake ledger — what bit, what bounced, and where the boundary is

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